Analytic Introduction to Logic

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Introduction

If you are reading this text, it is because you are taking a course on *logic*. Most of us understand intuitively that logic has something to do with thinking correctly, but, because one of the goals of this course is to improve our capacity for precision and thoroughness in forming our concepts, we will not be satisfied with a vague and poorly defined intuition. The question that should immediately arise, therefore, is "What precisely is logic?" This introduction will attempt to give a precise and rigorous answer to that question; for that reason, it will need to introduce certain ideas from epistemology, metaphysics, philosophical anthropology. Although these are not actually part of subject matter of logic—the *material object*, as we shall learn to call it—they are nevertheless indispensable for learning what logic actually *is*. Introducing these topics will also help serve one of the reasons why logic is taught at the beginning of philosophical studies at all: namely, to lay the groundwork for the study of metaphysics.

We will divide this introduction into three chapters: first, an introduction to the theory behind sciences and arts so as to show that logic can be classified as both; second an introduction to some basic concepts of philosophical anthropology, so that can better understand what an *habitus* (such as a science like logic) is; and finally a detailed look into the precise definition of logic.

Chapter 1

Logic as a Science and an Art

1.1 Sciences vs. Disciplines

Before you read this introduction, chances are that you thought of logic chiefly as a *discipline*: that is, as a body of knowledge that can be acquired through study. That is not wrong, exactly; however, it is limiting. This manner of looking at logic considers it only in the abstract—that is, it ignores the *being* or the *existence* of logic as it is actually found in reality. (In the logic of concepts, we will discuss the difference between *abstract* and *concrete* concepts.) In fact, this abstract way of looking at a subject is likely how you have looked at *all* academic fields, whether that be biology, quantum physics, astronomy, psychology, economics, or any other course of study. This is a somewhat natural reduction to make, because we generally gain access to such fields through formal education in a university setting: that is, by taking classes, reading books, writing papers, doing laboratory assignments, and the like.

In reality, however, a field of study does not reduce to an abstract discipline, a set of propositions that one must acquire. Rather, acquiring knowledge in a given discipline produces a real change in the mind—or to use more precise terminology, in the *intellect*—of the person who learns it. Let me illustrate with an example. Consider the following equation:

$$i\hbar\frac{d}{dt}\left|\Psi(t)\right\rangle ;=; \mathcal{H}\left|\Psi(t)\right\rangle$$

Readers familiar with quantum mechanics will recognize this as the time-dependent form of the Schrödinger equation.[1] For unfamiliar readers, it is doubtless completely unintelligible. In short, our human capabilities alone—in this case our *intellect*, our

[1] The equation describes the so-called wave function, which is used to determine the probability that a given particle is at a given location. Perhaps the most familiar use case is for describing the orbitals of electrons about an atomic nucleus.

capacity for knowledge of things in themselves—are insufficient in order to interpret an equation of this kind. Rather, in order to understand it, we would need to undergo a stable change in our intellect. The intellect alone and unaided will not serve, but it requires a modification or ordering—or, to use the technical term, an *habitus*.[2] Since the intellectual *habitus* in question helps us primarily to know things, rather than to do things, we will call it a *science*.

Notice that I have not identified the *science* of quantum mechanics primarily with the *discipline*. Rather, the *science* of quantum mechanics refers to the permanent and stable modification of the intellect, the *habitus* that is acquired by studying quantum mechanics. This is not the usual, modern usage of the term *science*, but rather, it reflects how classical philosophers used it—or rather, ἐπιστήμη (*epistéme*)[3] for the Greek philosophers, and *scientia* for Medieval thinkers. We will follow the classical usage of the term *science* in this introduction: an intellectual *habitus* that permits or helps us to acquire knowledge. A *scientia* is a real, existing, concrete attribute of a person—not merely an abstract body of knowledge, or discipline.

As you can probably tell from the foregoing discussion, logic is also a science, an ἐπιστήμη: not merely an abstract set of rules, but an actual intellectual *habitus* that exists (in technical terms, we say that it is *inherent*) in the logician's very person. It will, therefore, be useful to explain in greater detail what a science (an ἐπιστήμη) is, and how it is different from other *habitus*[4]—and also, how to demonstrate that logic is, in fact, a science.

1.2 The Essential Characteristics of any Science

Taking its cue from Aristotle's seminal work on epistemology, the *Posterior Analytics*, Medieval thinkers have generally described science (ἐπιστήμη) as *cognitio certa per causas*:[5] in other words, what distinguishes a science from mere opinion (Greek: δόξα, *dóxa*) is that one is able to link the individual facts known to their causes.

For example—one that is perhaps more familiar to readers than quantum mechanics—consider the example of medical science. For millennia, humans have been aware that certain diseases are contagious. Perhaps the most notorious example is during the

[2] I am going to use the Latin term *habitus* so as to avoid confusion with the English word *habit*, which primarily refers to an involuntary "tic" or pattern of behavior. An *habitus* (pronounced with a silent initial *H*) is a permanent disposition or ordering of something to an end.

[3] This term, of course, is the root of the word *epistemology*, the study of science or knowledge.

[4] The plural of the Latin word *habitus* is also *habitus*.

[5] See Aristotle, *Posterior Analytics* (henceforth cited as *PA*), tr. by G.R.G. Mure, I.2, 71b9-11: "We suppose ourselves to possess unqualified scientific knowledge of a thing, as opposed to knowing it in the accidental way in which the sophist knows, when we think that we know *the cause on which the fact depends*, as the cause of that fact and of no other, and, further, that the fact could not be other than it is."

Black Death in Europe (1347-1351): in order to avoid the spread of the disease, those who were infected, or were suspected of being infected, were often quarantined[6]—that is, isolated from the rest of society for forty days. In other words, Europeans had formed the correct *opinion* regarding the tendency of the plague to spread from host to host. However, it was not until the advent of the microscope and the development of the germ theory of disease (based largely on the work by Louis Pasteur and Robert Koch in the late nineteenth century) that the true *causes* of the plague became known—and thus effective prevention and treatment could be developed. We could say that the study of infectious disease did not truly become a *science* until the causes were discovered. Similarly, even today, a person might be aware that washing one's hands to avoid the spread of infection is a good idea, or that antibiotics are an effective treatment against bacterial diseases such as the plague, and yet not know that infection by bacteria is the *cause*. In that case, the person has the correct δόξα, or opinion, regarding these matters, but he does not posses ἐπιστήμη.

Similarly, there is the example of chemistry. For many years, scientists were aware that certain chemical species could only combine with one in other in certain proportions. For example, combining one volume of elemental hydrogen with another volume of elemental chlorine would produce, once the mixture had cooled to its original temperature, two volumes of hydrogen chloride (hydrochloric acid). However, combining two volumes of hydrogen with one volume of chlorine would produce a three-volume mixture of 1/3 hydrogen and 2/3 hydrogen chloride. The reason why the hydrogen remained a separate species was a mystery, until the atomic theory was proposed, and the existence of molecules (stable combinations of atoms) was discovered. In fact, elemental hydrogen and chlorine are both constituted by diatomic (two-atom) molecules (H_2 and Cl_2), and when they combine, they form molecules with one hydrogen and one chlorine atom (HCl). This arrangement explains why some hydrogen remains unreacted: once the chlorine is entirely consumed, it can no longer react. In other words, the phenomena observed in the laboratory were not satisfactorily explained until their *causes* were discovered.

Although the seeking of causes is what distinguishes science from opinion, Aristotle identifies three other essential characteristics of a science: namely, it must be *certain*—that is, it must lead to true knowledge without fail or error[7]; it must study that which can be attributed to its object in a *necessary* way (i.e., it identifies what cannot possibly be otherwise)[8]; and it must be *universally* applicable to its object of

[6] From the French *quarantaine* ("fortyish"), which is ultimately derived from the Latin *quadraginta*, forty.

[7] As Aristotle points out in the text cited above, by seeking the causes of a phenomenon, the whole purpose is to ensure that "the fact could not be other than it is" (*PA* I.2, 71b11), and as a consequence, such knowledge is free from error.

[8] See *PA* I.2, 71b14—15: "Consequently the proper object of unqualified scientific knowledge is something which cannot be other than it is."

study.[9]

Again, it can help to think of some examples. Consider the science of chemistry, which helps us to understand the properties of material entities, as regards the interactions that occur on the atomic and molecular level. It is clear that chemistry provides a great degree of *certainty*: if I mix sodium bicarbonate (H_2CO_3) with hydrochloric acid (a solution of HCl in water), I can be *sure* that table salt (sodium chloride, NaCl) and carbon dioxide (CO_2) will be produced. Similarly, one way to test to see if I have a base, such as sodium bicarbonate, is to mix a little hydrochloric acid into it; if it bubbles, I can be *certain* of it. In a similar vein, when I mix the acid into the bicarbonate, the outcome is the production of salt and carbon dioxide—and this result could not possibly be otherwise; that is, the evolution of these products is a *necessary* result. Finally, the result is *universal*, because it applies to *all* similar situations: in short, bicarbonate by nature reacts with hydrochloric acid, therefore *all* samples of bicarbonate will react with *all* batches of the acid. Another way of saying this is that chemistry studies only that which can be attributed *per se* (in and of itself) or *essentially* to the entities studied. Finally, as I mentioned above, chemistry is a science precisely because it seeks to identify the *causes* of these properties and reactions, which are to be found fundamentally in the atomic and molecular structures of the materials, especially in the arrangement and interaction of the electrons about the nuclei of the atoms.

The degree of certainty and necessity—and thus its universal applicability—can vary from science to science. Clearly, the degree of certainty and necessity that can be attained in pure mathematics is much greater than what can be obtained in a social science such as psychology or economics. Nevertheless, even in cases in which complete certainty, necessity, or universality are unattainable, a science provides useful knowledge *to the degree* that it is certain, necessary, and universal. For example, economics affirms that by and large (but not necessarily in every single case), when the price of a good or service rises, the quantity demanded falls. Although economics is unable to predict what an individual consumer will do, it does accurately account for general trends among many consumers. What a science cannot do is account for phenomena that have no essential link to their causes whatsoever; that is, science cannot account for what is *per accidens* (completely incidental). To a geneticist studying fruit flies, for example, what interests him are the laws the govern the inheritance of genetic characterists—what the flies happen to eat for food does not interest him, as such, since it does not affect heredity.

In summary: a science or ἐπιστήμη is an intellectual *habitus* that helps the intellect to understand a given object of study. As such, it always has four essential

[9] See *PA* I.11, 77a5-9: "So demonstration does not necessarily imply the being of Forms nor a One beside a Many, but it does necessarily imply the possibility of truly predicating one of many; since without this possibility we cannot save the universal, and if the universal goes, the middle term goes with it, and so demonstration becomes impossible. We conclude, then, that there must be a single identical term unequivocally predicable of a number of individuals."

characteristics:

1. Most fundamentally, it seeks to connect the phenomena it studies to their underlying causes. The seeking of causes is what distinguishes science (ἐπιστήμη) from opinion (δόξα), and the other three characteristics of a science all follow from this one.

2. It provides certain knowledge of its object—that is, it enables the one who possesses the science to know without error.

3. It accounts for phenomena that cannot be different from the way they are, or cannot come about in a different way. That is, it accounts for phenomena that are linked necessarily (*per se*), and not merely incidentally (*per accidens*), to the object of study.

4. Finally, it applies universally to the object of study: there is no science, as such, of singulars and particulars.

A couple of things follow from this. First of all, in order to determine whether a given intellectual *habitus* is a science, it is sufficient to show that it has these four characteristics. We will revisit this topic below. Second, it is not possible for science or ἐπιστήμη, in the strict sense of the term, to be false; that is, false information is the result, not of science, but of erroneous opinion (δόξα).

1.3 Speculative vs. Practical Sciences

We are not, however quite through with our examination of science: as it turns out, sciences come in different kinds, depending on their object of study and method. By now, you may have intuited that by defining science as the ancient and Medieval philosophers did—as *scientia* or ἐπιστήμη, that is, as an intellectual *habitus* whose purpose is to know the properties of a given object of study according to its causes—we are including under the banner "science" many more fields than are customarily placed there. By this definition, "science" is not limited to empirical fields such as biology, chemistry, or physics, but rather includes areas of study as diverse as mathematics, metaphysics, theology, and ethics. It is sufficient that each field have a specific and unique object of study (the *genus subiectum*) and appropriate methods for it.

Sciences can be categorized according to their finality—a criterion that is particularly useful for us. In the modern parlance, we speak of *pure sciences* such as biology, chemistry, and physics, and contrast them to *applied sciences* such as medical science and engineering. This division can be generalized to apply to all sciences (in the classical sense, as we have defined the term here): that is, some sciences seek knowledge of their object for its own sake, whereas other sciences seek the applications of the knowledge gained from them to some action. In the classical terminology, the former are called *speculative sciences*, whereas the latter are called *practical sciences*.

Again, it can be helpful to look at some examples. Consider the sciences listed in Table 1.1. What the speculative sciences listed in the left-hand column have in common is that they all seek knowledge of the object they study for its own sake. For example, a biologist can study the behavior of nesting robins without having any practical purpose whatsoever. This or that scientist may in fact have ulterior designs (he may be thinking of starting a business raising robins for their eggs), but that ulterior motive (or lack of one) does not change the character of his work *as a biologist*. The same can be said for chemistry, physics, mathematics, and metaphysics[10].

Table 1.1: A few examples of speculative and practical sciences. As much as possible, each practical science is placed next to the speculative science it is derived from.

Speculative	Practical
Metaphysics	—
—	Ethics
Biology	Medical science
Chemistry	Chemical engineering
Physics	Mechanical engineering
Mathematics	—

On the other hand, the practical sciences listed in the right-hand column all have a specific kind of operation (that is, a kind of action), in mind. What they produce is still a kind of *knowledge*, but in contrast to speculative sciences, for practical sciences, knowledge is not an end in itself—it is a means to action. For example, studying physics by itself will not be enough to tell you how to build a bridge; for that, you would need to study mechanical engineering. You need to learn how to apply the general principles discovered in physics to the more concrete problem of how to build structures. Similarly, studying biology—even human biology—by itself is insufficient to discover how to cure disease; that is the task of medical science. As you can probably tell from these examples, many practical sciences can be pared with a speculative science from which they are derived. In a similar vein—although it may not be obvious at first—ethics, the study of human actions (actions that involve the free choices made by the will) in order to discern whether they are right or wrong—is also in the column of practical sciences. This is so, because the entire purpose of ethics is to direct our

[10]In fact, metaphysics, the study of being (that is, of *ens*, that which is) as being, or being *qua* being, is rightly considered the most speculative of all sciences, since it has no direct practical application.

actions: we wish to discover which behaviors are compatible with human happiness, and which ones are not—in other words, the knowledge obtained through ethics is a means to the end of behaving well.

1.4 Is Logic a Science?

It is now time to turn our attention once more to our topic—logic—and to apply the principles we have been discussing. As we saw above, logic is clearly an academic discipline and field of study (at least when taken in the abstract), but the question now arises: is logic a *science* like biology, chemistry, and mathematics? In order to answer this question, we will need to apply our definition: a science is a *cognitio certa per causas*.

Is logic a *cognitio*—that is, an *habitus*, or stable disposition, of the intellect? In other words, in the absense of logic, would the intellect be impeded from acquiring certain types of knowledge? We will see this more clearly when we give the definition of logic below, but even based on our intuitive understanding of logic, we can confidently answer "yes": in fact, in the complete absense of logic, we would be unable to acquire any knowledge *at all*—and a systematic study of logic makes our thinking easier, more ordered, and less prone to error.

Does logic provide sure and certain knowledge by linking what it studies to its causes? As we will see, the answer is again "yes." Logic seeks the *causes of knowledge*: for example, through logic, I can be sure that conclusion C follows from premises A and B. This type of causality is different from that sought, say, in the sciences I listed above, all of which seek the *causes of being*, rather the causes of knowledge. That is, the *logical* causality proper to logic is distinct from the *ontological* causality proper to other sciences—but it is still causality.

Finally, what about the other essential characteristics of a science? Does logic provide *certain* knowledge—that is, knowledge without error? Does it supply knowledge that follows necessarily from the nature of its object of study? Does the knowledge it furnishes apply universally to its object? Again, the answer to all three questions is in the affirmative. As regards certainty, the very purpose of logic is to facilitate our reasoning ability and make it less prone to error. As regards necessity, again, logic provides the tools we need to see that the conclusions we reach are the correct ones, and could not be otherwise. As regards universality, it is clear that logic applies to *all* of the works of our intellect, without exception.

In fact, we should carefully define our terms, here, because there are two distinct entities without which knowledge is simply impossible: there is our intellect, which is the faculty that permits us to make acts of knowledge—knowledge that captures the very essence and being of things; and there is the intellectual *habitus* that we are discussing here. For this reason, intellect, which is sometime called our *reason* (in Latin:

ratio; in Greek: λόγος or *lógos*)[11], can be called a kind of "natural logic." However, logic, in the strict sense of the term, refers to an *habitus*, an acquired ability that *helps* the intellect in its operation. However, it should be noted that this logic-as-science, or scientific logic, can be acquired in an imperfect way (that is, the intuitive grasp of logic that we acquire from the mere use of our reason), or else in a "perfect" or systematic way—which is what this course is attempting to do.

We must, therefore, conclude that logic is indeed a science.

1.5 *Logica utens* vs. *logica docens*

Based on our previous discussion, the another question arises: is logic a *speculative* science or a *practical* science? As we saw, the purpose of a speculative science is knowledge for its own sake. Logic certainly seems to fit the bill. St. Thomas Aquinas in his commentary on Aristotle's *Posterior Analytics* describes the purpose of logic in this way:

> [A] certain art is required, one that can direct the very act of reason, through which man, if you will, in that same act of reason may proceed in an orderly manner, more easily, and without error, and this art is logic, that is, the rational science.[12]

Thus, the purpose of logic is to order the individual actions of our intellect in such a way that the acquisition of knowledge is easier and less error-prone. When someone has acquired the science of logic, it will be much easier for him to distinguish truth from falsehood; in fact, we could summarize Aquinas' description here as saying that logic is our instrument for discovering true knowledge, and ensuring that it is *true*.[13] As such, we can categorize logic as a speculative science, albeit a unique one, because (as we will see) its ultimate goal is not knowledge of its object—the works of reason—but rather knowledge of the objects of all the *other* sciences. That is, we do not study

[11] In the strict sense, our *reason* refers to our capacity for acquiring new knowledge starting from knowledge that we have already acquired: that is, our ability to infer or draw conclusions, or, to use the technical term, our *ratiocinium*. However, when taken broadly, the term *reason* is synonymous with the intellect.

[12] Thomas Aquinas, *Expositio libri Posteriorum Analyticorum* (henceforth cited as *EPA*), proemium 1-2: "[A]rs quaedam necessaria est, quae sit directiva ipsius actus rationis, per quam scilicet homo in ipso actu rationis ordinate, faciliter et sine errore procedat. Et haec ars est Logica, idest rationalis scientia" (my translation).

[13] A reminder that, for the ancients, *knowledge* (*scientia*, ἐπιστήμη) as such is always true. There can be false *opinion* (δόξα), but not false *knowledge*; we will follow that convention in this course. (I will also mention that the ancients used the same term *scientia* or ἐπιστήμη to refer both to the *habitus*—that is, the habitual knowledge that is not necessarily in act—and to the individual *acts* of knowledge that flow from the *habitus*.)

logic in order to study the works of reason for their own sake; rather, we study the works of reason in order to *use* them in biology, mathematics, physics, engineering, and every other science (including logic itself)—indeed in every act of the intellect. We can summarize this idea by saying that logic is *instrumentally speculative*: it certainly seeks knowledge for its own sake, but it does so strictly *as a means or instrument* of obtaining that knowledge.

A new question immediately arises: if logic is a means to obtaining knowledge, and obtaining knowledge is an action, doesn't that make logic a practical science? It does indeed. What this means is that—somewhat surprisingly—"speculative" and "practical" are not mutually exclusive: a science can be both.

How is this possible? It goes back to what we touched on earlier: a science, being an intellectual *habitus*, makes it possible for the intellect to do certain actions that it was unable to do before. An *habitus* is only required when there is a special difficulty in acquiring the knowledge in question: for instance (referring to the example above), I only need to learn (that is, acquire habitual knowledge of) quantum physics if I need to understand things like the Schrödinger equation. As we discussed above, many speculative ("pure") sciences can be pared with a practical ("applied") science—for instance, mechanics (physics) with mechanical engineering, or human biology with medical science. Why is a distinct applied science necessary? In short, because the application is not always obvious: mechanics, for example, tells me how to calculate the forces acting on a rigid body; it does not tell how to apply this knowledge to building a bridge. Similarly, human biology tells me that cancers are caused by genetic mutations that accumulate in the DNA of the body's cells; it does not tell me the best *treatment* for removing that cancer. In sum: when the application of speculative knowledge poses a particular difficulty, a distinct *habitus* (in this case, a distinct *science*) is required in order to make that application.

With logic, however, the situation is a bit different. Since logic studies the very works of our reason and (as we will see) the proper construction of relations between them, merely studying the object of logic immediately hones our ability to reason. We will return to this idea below.

In any case, it is clear that, in addition to being a speculative science, logic is also a practical science: although logic is speculative insofar as its end goal is knowledge, its manner (or *mode*) of achieving that goal is practical: said in other terms, logic teaches us *how to know*. For this reason, logic is said to be a *modally practical* science.

1.6 Is Logic Also an Art (τέχνη)?

Attentive readers will note that Aquinas, when describing the goals of logic, does a curious thing: he calls it an *art* (*ars*; in Greek: τέχνη or *téchne*). That is surprising, because we have just spent the last few pages demonstrating that logic is a *science*. It may also surprise readers that Aquinas calls it an "art" along with painting, music, sculpture, and theater. In order to understand this, it is best to take a moment to understand what Aristotle and Aquinas meant by *art* or *τέχνη*.[14]

For Aquinas, an art could refer to any intellectual *habitus* that results immediately in action: in the classical formulation, *recta ratio factibilium*, right reason applied to things that are made. In other words, it is a use of reason or the intellect (*ratio*) that is properly ordered (*recta* or "right") to the accomplishment of determined operations or actions (*factibilium*).

Again, some examples are helpful: suppose you are given a piece of marble, and instructions to carve it into a statue of David. For the vast majority of readers, the result would most likely not look anything like Michelangelo's famous statue of David. As we saw before with the sciences, in order for the sculptor to be able to set chisel to stone and produce a work of beauty, he must first acquire an *habitus*—in this case the art of sculpture. I think it is clear that the same reasoning applies to a violinist, an opera singer, a painter, and a photographer. However, for Aquinas, *art* is not limited to what we would call the "fine arts"; it applies to any *habitus* that is necessary for skilled action. Therefore, surgeons, architects, athletes, lawyers, carpenters, writers, orators, accountants, and entrepreneurs are all "artists" by this definition.

Notice that arts can easily be distinguished by the kind of action they produce. Some arts, called the *mechanical arts*, produce something that is external: for example, the art of painting results in physical artwork such as a fresco or canvass, and carpentry results in woodwork such as tables and chairs. On the other hand, the *liberal arts* produce something that is internal to the intellect: for instance, the art of practicing law does not leave a physical work, but rather produces intangibles such as legal precedents, decisions, motions, and verdicts.

As with the sciences, arts have certain characteristics in common. Let us take, for example, the art of carpentry. A carpenter takes a raw material—in this case, wood—and fashions it into finished products, such as tables and cabinets. In order to do so, he applies the techniques that he has learned: how to cut, trim, and smooth out the wood, putting the pieces together and gluing them in the right way, adding the right kind of finish, and so on.

We can generalize this idea to every art: the practitioner of an art takes some indeterminate entity—its "raw material"—something that *can* be fashioned as desired, but *need* not be. He transforms it using means that are determined and universal, so as to produce the desired product. That is, we could say that, in general, an artist's task is to take an indeterminate "matter" and remove its indetermination. For instance, when Michelangelo Buonarotti was given blocks of marble, the number of possible statues he could make from them was infinite; it was fully *indeterminate*. However, the more

[14] From the Greek word *τέχνη*, we derive a number of English words, including *technique* and *technology*.

he worked on them, the closer they became to being fully *determinate*: say, a Pietà.

In fact, any *habitus* with these two characteristics can be classified as an *art* or *τέχνη*. With the mechanical arts it is easy to see, but the same holds for all of the liberal arts: a defense lawyer must take all of the facts of the case and his knowledge of the laws to formulate a cogent defense for his client; an orator must take the infinite possibilities of a language and formulate a beautiful speech; a composer must take tones, rhythms, dynamics, and instrumentation, and form a musical composition.

So, what about logic? Could logic be classified as one of the arts? To summarize what we saw just now, an *habitus* is an art if it fulfills the following characteristics:

1. It takes an indeterminate matter that does not infallibly attain its fully determined form.

2. It transforms that matter into a finished product using determinate and universal means.

Does logic qualify? (1) Regarding the first condition, as we shall see, the task of logic is to take the works of our reason (concepts, enunciations, and arguments) and to construct relations between them according to very strict and determined rules. Even though our reason is ordered to the truth, there is nothing preventing us from getting the relations between them incorrect: that is, we can fall into error and falsehood. Therefore, these works of reason constitute the *indeterminate matter* of logic. (2) Regarding the second condition, among all of the possible relations among the works of our reason, logic constructs—using *universal and determined rules*—the *correct* relations, the ones that will produce *true* knowledge. It follows that logic is indeed an art: a liberal art, evidently, as it does not produce a external product.

1.7 How Logic Manages to Be All Three at Once

Ordinarily, speculative sciences, practical sciences, and arts can be clearly distinguished from one another, even when their subject matter is closely related. Continuing the example I mentioned above, human biology is the speculative science at the root of all medical matters; medical science is the practical science that applies the knowledge of human biology to the treatment of disease; and medicine is the art that applies medical science to concrete acts of treatment. Three distinct *habitus* are at play here, because neither of the two applications is trivial. A surgeon, for example, can hardly expect to be sufficiently trained simply by acquiring the speculative science of human biology. In fact, not even detailed knowledge of medical science is enough: he can spend years learning *how* surgeries are done, and yet be completely unprepared for performing an actual surgery. In short, he does not become a surgeon without first acquiring the art of surgery—which in this case implies not only speculative and practical knowledge, but also a training in the coordination of the hands. In other words, what renders the three *habitus* distinct from one another is that there is, if you will, a considerable gap to close between the speculative science and the practical science, and between the practical science and the art.

With logic, however, we use our intellect to study works produced by that same intellect. Therefore, the distance that is present in other fields is absent in logic. We have already noted above that the speculative aspect of logic, as a science, cannot be really distinguished from its practical aspect. It is also clear from our discussion of arts that the *art* of logic cannot be really distinguished from the *science* of logic. In other words, formal, scientific instruction in logic—what is traditionally called *logica docens*[15]—being the study of an object that is intrinsic to the very intellect that studies it, results immediately in an improvement in the exercise of logic—what is traditionally called *logica utens*[16]. Later on in the course, you will learn to distinguish and relate different kinds of concepts, how to relate enunciations to one another, and how to form and analyze syllogisms. You will be able to use those skills immediately; there is no need for an *additional* science, nor for a distinct *art*, in order to apply them.

1.8 How Logic is Necessary

Before closing this chapter, I would like to say a word about the manner in which logic is a necessity. As Aquinas says, "the necessary is what which cannot not be."[17] In what way can logic be said to be necessary? Let us take a moment to unpack this idea further.

Aquinas argues that there are fundamentally four kinds of necessity:[18]

> In fact, one way that [being necessary] pertains to something is by an intrinsic principle, which can be either material, such as when we say that it is necessary for everything composed from contraries to be corrupted, or formal, such as when we say that it is necessary for the three angles of a triangle to be equal to two right angles. This is natural or absolute necessity.
>
> Another way that something is unable not to be is by an extrinsic principle, either an end or an agent. For instance, and end is necessary when something cannot be obtained without it, or else cannot be obtained well without it, as food is necessary to life and a horse for a journey. This is

[15] From the Latin *doceo*, "to teach."

[16] From the Latin *utor*, "to use or employ."

[17] Thomas Aquinas, *Summa Theologiae*, Leon. vols. 4–12, Typographia Polyglotta S.C. de Propaganda Fide, Rome 1888–1906 (henceforth cited as *S.Th.*), I q. 82, a. 1, *responsum* (my translation).

[18] Attentive readers familiar with Aristotle will observe that each kind of necessity is closely tied to a corresponding species of *cause*: agent, formal, material, and final.

> called the necessity of end, which is also sometimes called utility. There is necessity regarding agents when someone is forced by an agent, so that he is unable to do the contrary. This is the necessity of coercion.[19]

Let us look at each of these in turn:

- A *material* necessity has to do with the limitations that matter places on material beings: for example, fact that they take up space, must exist in a single location, and are corruptible.
- A *formal* necessity (also known as a *natural* or *absolute* necessity) stems from the substantial or accidental forms of an entity. As you will learn in the course, this kind of necessity can be of two kinds: it can stem directly from the entity's essence, and thus be part of it's very definition, as when we say that it is necessary for man to be rational; or else it can be a property that stems from that essence, as when we say that man is capable of laughing.
- A *final* necessity—which is probably the kind of necessity you are most familiar with—refers to the means required in order to obtain an end. It is what we generally refer to when we say that we "need" something. Aquinas observes that this kind of need can be *strict* or *moral*.[20] Food is an *strict* necessity for life. Taking a car to go somewhere 10 miles away is not a strict necessity, but unless you really enjoy walking, taking a car is much better. Thus it is a *moral* necessity.

What are we to say about the necessity of logic? First of all, as we saw, logic is essential for obtaining knowledge. *Natural logic*, which is another name for reason itself, is, according to our analysis, a *strict* necessity for knowing anything, as should be evident. As for obtaining systematic, scientific knowledge, it should be noted that such knowledge can be obtained in an imperfect way (in which we understand only the easiest conclusions of the science and our dominance of the material is fuzzy) or in a "perfect" way (in which all of the conditions of possessing a science are satisfied, with the necessary rigor). In order to get an imperfect grasp of a science, natural logic is sufficient—although having a scientific logic is much better. In a word, scientific

[19] THOMAS AQUINAS, *Summa Theologiae*, Leon. vols. 4–12, Typographia Polyglotta S.C. de Propaganda Fide, Rome 1888–1906 (henceforth cited as *S.Th.*), I q. 82, a. 1, *responsum* (my translation).

[20] Don't be confused by the term "moral": it is taken in an etymological sense from the Latin term for custom (Latin: *mos*, *moris*). The idea is, people are accustomed to using suitable means that are not *strictly* necessary.

As regards ethical norms, they are in fact necessities of end, because following them is a necessary condition for happiness. Nevertheless, not all of them are *strict* necessities. following the basic rules of human decency, such as obeying the Ten Commandments, is indeed a *strict* necessity for obtaining happiness (that is, *not* following them is incompatible with happiness); however, some actions—such as entering religious life for those who are called to it—are only necessary for *greater* happiness. The latter are "moral" necessities in the sense we are taking the term.

logic is a *moral* necessity of end for this kind of knowledge. However, scientific logic is a *strict* necessity of end for "perfect" scientific knowledge.

In sum: we conclude from the foregoing discussion that logic is simultaneously an instrumentally speculative science, a modally practical science, and an art, whose object is the works of our reason. Natural logic is a strict necessity for all knowledge whatsoever. Logic studied scientifically is a great help for obtaining scientific knowledge, and absolutely essential for "perfect" scientific knowledge. What remains to discuss in greater detail, therefore, is the *object* of logic, which we will do after a brief discussion of some key concepts in philosophical anthropology that are helpful for understanding what we have just gone over.

Chapter 2

A Primer on the Metaphysics of Man

In this chapter, we will explore some of the essentials of philosophical anthropology: the philosophical, or metaphysical, study of mankind.[1] We will do this for two reasons. First of all, although this course is designed primarily to study logic, it is also intended to be the reader's first exposure to some of the terminology and ideas that will form the basis of their philosophical inquiry: for example, *ens*, *substance*, *accident*, *faculty*, *habitus*, and *operation*. Second, by laying the philosophical bases, it will be much easier to understand the definition of logic, which is our ultimate goal.

2.1 What are metaphysics and anthropology?

Before beginning in earnest, since this is likely the reader's very first exposure to the sciences of metaphysics and philosophical anthropology, a short introduction to what we are talking about is in order.

Metaphysics is the study of anything that exists—of *entities* (in Latin: *entia*), as we will call them—not according to this or that particular aspect, but simply *because they are*. That is, according to Aristotle's famous definition, metaphysics is the study of existing things (*entia*) *qua* being.[2] We saw above that sciences study their object with

[1]We call it *philosophical* anthropology to distinguish it from *cultural* anthropology, which is empirical, rather than philosophical, in nature.

[2]As Aristotle states in *Metaphysics*, "There is a science which studies Being *qua* Being, and the properties inherent in it in virtue of its own nature. This science is not the same as any of the so-called particular sciences, for none of the others contemplates Being generally *qua* Being; they divide off some portion of it and study the attribute of this portion, as do for example the mathematical sciences. But since it is for the first principles and the most ultimate causes that we are searching, clearly they must belong to something in virtue of its own nature" (Aristotle, *Metaphysics* [henceforth cited simply as *Metaph.*], tr. by W.D. Ross, Oxford, Clarendon Press, 1924, Γ, I.1, 1003a21-28).

In languages, like English, that do not distinguish well between *being* in the sense of "something that exists" (in Latin: *ens*) and *being* as a synonym of "to be" (in Latin: *esse*), this definition is often translated the goal not only of understanding its properties, but with the hope of tracing those properties all the way back to their causes. As can be seen, metaphysics is the most general of sciences: no created entity is excluded from it whatsoever. Therefore, unlike other sciences, it seeks the properties that can be attributed to *all* entities without exception,[3] and it seeks the causes on which *all* things depend. It should be noted that metaphysics studies *created entities*, and thus God is actually *not* the object of this science. Rather, He is of interest as both the First and Ultimate extrinsic Cause of all entities:[4] the First Efficient and Exemplary Cause, and the Ultimate Final Cause.[5]

Aristotle called this science various names, including the "first Science," "theology", and "wisdom"—but not "metaphysics." That name was given first by the compilers of Aristotle's works located in Alexandria, Egypt, simply because they placed those treatises *after* Aristotle's works on physics (in Greek: μετὰ τὰ φύσικα, *metà tà phýsica*). Despite the trivial origin of the name, by an accident of the Greek language, it is actually quite appropriate: μετά can also mean "beyond," and it turns out that metaphysics is the only philosophical discipline that can transcend *beyond* the physical (that is, the material) world.

When metaphysical principles and findings are applied to mankind, the resulting science is *philosophical anthropology*, which could be defined as the study of man, not merely regarding this or that aspect of man, but *insofar as he is*, or *insofar as he is man*. Like other applications of metaphysics, philosophical anthropology seeks to find the properties that apply universally to its object, man, and links them to their ultimate intrinsic and extrinsic causes (that is, man's constitutive metaphysical principles, and God as First and Ultimate extrinsic Cause.) For our purposes, it will be enough to look at some of the intrinsic causes of man—that is, some of the metaphysical principles that constitute him.

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as "being *qua* being" or "being as being." However, this rendition is rather misleading: metaphysics studies existing *things* (*entia*) insofar as they are, not primarily that perfection or attribute of existing things that we call "being" (*esse*).

[3]We will touch on some of these, as they regard logic, when we go over the *transcendentals* in the part of the course on the logic of the concept.

[4]Thus, God is not a mere entity, in the strict sense of the term, not a mere thing that exists: rather, He is Being Itself (Latin: *Ipsum Esse*).

[5]This pattern—that the cause is not the object of a science—is common throughout all sciences. For example, consider biology, whose object is living things as regards their bodily life. Biology links the life processes of living things to their chemical causes: for example, the chemical pathways by which organisms extract the energy in the food they eat helps to explain their diets and eating habits. Biology, as such, studies the *organisms* and their properties, not the chemical causes of those properties; however, of course it relies a lot on the findings of chemistry to understand those properties in greater depth. In a similar way, the task of metaphysics is to study created entities, not God; but knowing something about God helps the science considerably in its understanding of entities.

2.2 Entities (*Entia*), Substances, and Accidents.

The first notion that we will introduce is *ens* (or its plural in Latin *entia*), which is the term that classical and Medieval philosophers used to translate Aristotle's τὸ ὄν, the present participle of the Greek verb εἰμί ("to be"). In English, we will be translating *ens* as "entity." As we discussed above, an entity is simply something that is, or (in modern parlance) exists[6]: as Aquinas puts it, *id quod est*, "that which is." Aquinas actually employs an even more precise description: *id quod habet esse* ("that which has being"), which emphasizes the fact that the term "being" (*esse*) refers to a perfection of the entity in question—a reality that is constitutive of that entity—not a mere logical fact.

It should be evident that, outside of God, who is a special case (He is not an entity exactly, but is best characterized as Being Itself, *Ipsum Esse*), anything that *is* at all, in any way, is an entity. That is, there are nothing but entities—anything else, simply is not (does not exist). But what counts as an *ens*? Let us suppose that there is an oak tree growing in the back yard. Since the oak tree clearly *is*, no one can doubt that it is an entity (*ens*). However, the oak tree does not merely "be," as if being could occur in isolation; it has *attributes* and properties. For example, it has a height, a width, a weight, a place, textures, colors, a time, and a configuration (among many, many others). Are these attributes themselves entities (*entia*)? The answer is not as obvious, since these attributes are so closely linked to their substrate, or underlying subject, the oak tree. Nevertheless, we can resolve the issue the way Aristotle did: since each attribute can be predicated of the oak tree using the verb *to be*, it is a sign that we intuitively grasp that these attributes enjoy a certain type of being. The height of the tree, for example, certainly exists (more precisely, it *is*), for if there *were* no height, it would be a very strange tree indeed. On the other hand, the attributes of the tree hardly enjoy the same kind or degree of being as the tree itself. The oak tree is independent, it is "by itself" and "on its own"—or as we say in metaphysical terminology, it *subsists*. In contrast, the attributes do not exist "on their own"—it would be absurd to think of height, color, weight, or any of the examples given above, without some material body to which they correspond—but rather they are always "in" another entity, or as we say, they *inhere* or are *inherent* in another entity.

By this reasoning, we have discovered the first and easiest division of entities: some entities, which we will call *substances* (Greek: αἱ οὐσίαι) are *subsistent*, that is, they exist

[6] In metaphysics, there is actually a difference in meaning between *existence* (Latin: *exsistentia*; in later Latin *existentia*, literally "a coming forth") and *being* (*esse*). Aristotle uses the terms τὸ ὑπάρχειν (*to hypárchein*) and τὸ εἶνια (*to eînai*) to render the same idea. *Existence* is the mere fact that something is, whereas *being* refers to the ontological perfection that renders existence possible. Although in modern parlance, the terms have come to be practically interchangeable, in Aristotle and St. Thomas Aquinas, they maintain their distinct meaning. Thus, as in most cases I am referring to being as a perfection, I will try to use the verb *to be* as much as possible when referring to entities.

by themselves. Others, which we will call *accidents* (Greek: τὰ συμβεβηκότα,[7] *ta symbebekóta*, literally, "things that stand alongside") are mere attributes or modifications of a substance—as we say, they are merely *inherent* in a substance. Although substance and accident can, in a sense, be thought of logically as the different "types" of entities, they are best thought of as constituent principles of a thing: the oak tree has an intrinsic principle that makes it an oak tree—the substance, or, in this context, more often referred to as the *essence*—and a set of distinct principles, also intrinsic, that give the tree its various perfections, such as height, weight, location, and the like.

At this point, it is appropriate to make a note about the term *substance* (Latin: *substantia*), which can take on slightly different, but closely related meanings depending on the context. At the beginning, when we distinguished substance and accident as different kinds of *entia*, we were considering substances as fully constituted wholes: for instance, an oak tree, including all of its perfections or attributes. Medieval philosophers sometimes used the term "supposit" (Latin: *suppositum*, Greek: ὑπόστασις, *hypóstasis*) to denote this meaning of the term *substance*. As we just saw, however, *substance* can also be thought of as an intrinsic constitutive principle that is composed with the accidents; Western European philosophers have generally used the term *essence* (Latin: *essentia*) to denote this usage. The two meanings are so closely related that Aristotle uses the very same word, οὐσία, for both.

Regardless of how we are using the term *substance*—either as supposit or as essence—it is always the answer to the question "What is it?" Only substances subsist, and thus, only they constitute a full answer to this question. As Aristotle clarifies, unlike the accidents, only substance constitutes what a thing is simply *because* it is (τὸ τὶ ἦν εἶνια, *to ti ên êinai*)—not because of some perfection or attribute.

In sum: all created realities (which includes everything with which we have habitual contact) are rightly called *entia* or entities, because they *are*. Substances are the only entities to *subsist*, or be "by themselves," and thus they are entities par excellence. Substances are constituted by an intrinsic principle, called the *essence*, that gives each substance its definition (that is, the essence makes that substance what it is), and the essence is composed with various perfections or attributes called *accidents*. Entities are always real; anything that lacks being entirely is *not* properly called an entity. (This idea will become important below, when we talk about *entia rationis*.)

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[7] Readers familiar with classical Greek will note that Aristotle's term is derived from the preposition σύν ("with"), used as a prefix, and the verb βαίνο "to step or walk"—hence the verb συμβαίνω. Although the verb in ancient Greek had the primary meaning "to stand with one's feet together," it also meant "to stand with or beside," which is the meaning that Aristotle is using here. The form τὰ συμβεβηκότα (singular: συμβεβηκός) is the perfect participle.

2.3 The Intrinsic Composition of Man: Faculties, *Habitus*, and Operations

For the moment, that is enough of an introduction to intrinsic metaphysical principles to begin to apply them to our subject of interest, which is man—who is the only creature that makes use of logic: the sub-human creatures are incapable of it, and angels do not need it. A human being is, like our oak tree, evidently something that *subsists*: a substance. And therefore, like any substance, he is composed of an *essence* (an ontological principle that makes him who he is) and various *accidents*, which are his perfections. His essence is itself composed of ontological principles that Aristotle calls *prime matter* and *substantial form*—in man, we know these as the *body* and the *soul*—a topic that you will study at length in cosmology (the philosophy of nature) and metaphysics.

For the purposes of our discussion about logic, it is sufficient to examine certain types of perfections: faculties, *habitus*, and operations.

Faculties

It should be obvious from experience that the substances we encounter do not merely "be"; they also "do"—that is, they perform a whole series of actions (which we will learn to call *operations*). Even inanimate objects at least exert forces (for instance, the force that a rock exerts when I stub my toe against it), emit or reflect radiation (which is how we can see things), and can be made to interact with other substances (for instance, think of all the chemical reactions that occur constantly). Evidently, as we go up the scale to nobler creatures, such as plants, animals, and humans, the number and perfection of the operations performed increases. For example, consider those substances that enjoy vegetative life:[8] they have available to them a whole series of operations that are unknown to inanimate objects, such as growth, nutrition, and reproduction. Animals go a step further: they display sense knowledge and appetite, and most animals can move from place to place on their own power. Finally, humans, aptly called *rational animals*, can do everything that plants and animals can do, but in addition, we are able to know things in themselves, and to love them for their own sake.

[8] In philosophy, we commonly refer to such substances as *plants*, but I would like to emphasize that in this context we understand the term in a philosophical, not a biological, sense; that is, corporeal substances that are capable of growth, nutrition, and reproduction, but do not enjoy sense or rational operations. Thus, these substances include many that are not in the plant kingdom as biology defines it, such as bacteria, archaea, fungi, and protists.

It may be less obvious at first glance, but, save in the case of God[9], the raw essence[10] of a substance is incapable of acting or doing anything on its own. In order for the substance to act in any way, it first needs a *capacity to act* in that way, and that capacity is itself an accident or perfection of the substance. For example, in order for an animal to move from place to place, it must first have a capacity for locomotion (which in higher animals entails a muscular system and a system of motor nerves). Similarly, our capacity to know things in themselves, which, as we saw, is called the *intellect*, is absolutely necessary in order for us to make acts of knowledge. In fact, all of our operations have their root in some intrinsic capacity: our physical movement, our senses, our memory, our imagination, our emotions, our intellectual knowledge, and our acts of love. In general, any capacity that produces an operation can be called an *active potency* or *active power*. If the active potency produces an operation that only an animal (rational or irrational) can do, we call it a *faculty*.

Table 2.1: Some operations and their corresponding faculties.

Operation	Faculty
Movement from one place to another	Locomotion
Sensation of material entities as regards color	Sight
Sensation of material entities as regards sound	Hearing
Recollection of past benefits and dangers	Sense memory
Composing coherent spatial representations	Common sense
Perceiving spatial relations, continuity, and benefit or harm	*Vis cogitativa*
Desiring the easy and pleasurable good	Concupiscible appetite
Desiring the difficult and arduous good	Irascible appetite
Knowing things as they are	Intellect
Loving things for their own sake	Will

Since this is a course on logic, it should come as no surprise that the faculty we are most interested in as the "cause" of logic is the intellect: our capacity to know things as they are. It is worth taking a moment to understand fully what we are taking about. We should note, first of all, that it is possible too *know* in different ways. At its most fundamental level, to *know* something means that the knower, in a certain sense, or a

[9] In Thomistic and Aristotelian philosophy, God is perfectly identical with both His essence and His operations. God has no accidents, or more precisely, every perfection that for a creature would be an accident, in God, is His very Essence.

[10] Remember, when we distinguish the *essence* from the *substance* in metaphysics, by "essence" we mean that ontological principle that serves as the subject or substrate of the accidents. (Although we have not yet discussed it here, is also the potential principle that receives the act of being. You will learn more about that in your course on metaphysics.)

certain part of him, actually *becomes* the thing known. We wouldn't normally think of it as a kind of "knowledge," but in a way the block of marble that Michelangelo formed into a statue of David came to "know" the model that was in Michelangelo's mind once the chiseling was complete. In a similar way, when a dog recognizes its owner, it does so by means of the *sensible image* that it has formed in its internal senses.[11] This image, in which the accidental forms of the owner have become impressed on the sense faculties of the dog, can be characterized in a much more proper sense as *knowledge*: we would call it *sense knowledge*, the kind of knowledge that can be acquired only through the internal and external senses, without the need for a properly rational faculty.

However, in logic we are not directly interested in sense knowledge. Through sense knowledge, an animal can only gain access to the object of its knowledge insofar as it corresponds to (or goes against) its appetites: in short, an animal knows something only *insofar as it is perceived as beneficial or noxious*. On the other hand, human beings are able to do something considerably greater: in addition to perceiving an object as beneficial or noxious, we can grasp *what* the object is *in itself*. A dog will wag its tail when its owner brings it food; it is idle to ask the dog *what food is* or *what an owner is*, still less whether these entities *exist in reality* or not, but a human being has no trouble doing either.[12] As a *rational* animal, a human being is capable of grasping two

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[11]We now know, through modern biology, that the internal senses are located in the central nervous system, especially the brain.

[12]C.S. Lewis in his book *That Hideous Strength* provides perhaps the best description of the difference between sense and rational knowledge that I have come across: "Mr. Bultitude's [a tame bear that figures prominently in the story] mind was as furry and as unhuman in shape as his body. He did not remember, as a man in his situation would have remembered, the provincial zoo from which he had escaped during a fire, nor his first snarling and terrified arrival at the Manor, nor the slow stages whereby he had learned to love and trust its inhabitants. He did not know that he loved and trusted them now. He did not know that they were people, nor that he was a bear. Indeed he did not know that he existed at all: everything that is represented by the words *I* and *Me* and *Thou* was absent from his mind. When Mrs. Maggs gave him a tin of golden syrup, as she did every Sunday morning, he did not recognise either a giver or a recipient. Goodness occurred and he tasted it. And that was all. Hence his loves might, if you wished, be all described as cupboard loves: food and warmth, hands that caressed, voices that reassured, were their objects. But if by a cupboard love you meant something cold or calculating you would be quite misunderstanding the real quality of the beast's sensations. He was no more like a human egoist than he was like a human altruist. There was no prose in his life. The appetencies which a human mind might disdain as cupboard loves were for him quivering and ecstatic aspirations which absorbed his whole being, infinite yearnings, stabbed with the threat of tragedy and shot through with the colours of Paradise. One of our race, if plunged back for a moment in the warm, trembling, iridescent pool of that pre-Adamite consciousness, would have emerged believing that he had grasped the absolute: for the states below reason and the states above it have, by their common contrast to the life we know, a certain superficial resemblance. Sometimes there returns to us from infancy the memory of a nameless delight or terror, unattached to any delightful or dreadful thing, a potent adjective floating in a nounless void, a pure quality. At such moments we have experience of the shallows of that pool. But fathoms deeper than any memory can take us, right down in the central warmth and dimness, the bear lived all its life" (taken from C.S. Lewis, *That Hideous strength*, New York, Scribner 1999, chapter 14, "Real

important aspects of the object of his knowledge that are unavailable to an irrational animal: its *essence* (*what* that object is) and its *being* (*whether* or not, in fact, that object is or exists). As we will learn, these aspects of rational knowledge form the basis for logic's object of study, which is the works of our reason. Through our concepts, we apprehend *essences*, and through our enunciations (and through the reasoning that we sometimes need to form enunciations) we make judgments regarding *being*.

The distinction between sense and intellectual knowledge cannot be overemphasized: the distinction is *not* made in a large number of philosophers, especially those who derive their ideas, directly or indirectly, from the Empiricist school. It is not my purpose to denigrate the valid insights brought forward by great philosophers such as John Locke, David Hume, and Immanuel Kant; however, I think each of these authors *can* be constructively critiqued for overlooking this important point. As you will learn in philosophical anthropology, our capacity for intellectual knowledge and love (our intellects and wills), as distinct from sense knowledge and sense appetite, is the most important evidence in favor of the *immateriality* of our souls. Our intellects do not depend for their existence on our bodies, but rather emanate directly from our rational, spiritual, immaterial souls. What confused the Empiricists, in my opinion, is that our intellects, emanating from a soul that informs a body, requires sensation as a necessary condition for its operation (barring a divine intervention). It is true, as the Scholastic saying aptly states, that *nihil est in intellectu quod prius non fuerit in sensu* ("there is nothing in the intellect that was not first in the senses"); nevertheless, intellectual knowledge is not *produced* by the body (as sense knowledge is), but rather by a spiritual, immaterial faculty that transcends the body. Said in other terms, the brain is not the "organ" that "holds" the intellect; rather, the intellect makes use of the brain (and the rest of the body's sensory apparatus) in order to function properly.

This is a good moment to make some more remarks about terminology. The most correct and proper name for the faculty through which we know things as they are is the *intellect* (Latin: *intellectus*; Greek: νοῦς, *noûs*). The term "mind" (Latin: *mens*) is less precise, because it can refer also to emotional states. Similarly, "consciousness" (a favorite of many contemporary philosophers) tends to imply that the person in question is actually *exercising* his intellect—i.e., that he is actively aware of his surroundings—whereas the intellect exists even when the person is asleep or otherwise unconscious. (Indeed, it exists even before he has developed the use of his reason.) Moreover, "consciousness" and "self-awareness" are difficult to define, and could easily be attributed to some of the higher animals. Finally, you will often encounter the term "reason" (Latin: *ratio*; Greek: λόγος, *lógos*) used as a rough synonym for the intellect. When speaking in a broad sense, it is fine, but strictly speaking the term "reason" applies to our ability to take known premises and derive previously unknown conclusions

Life is Meeting," part III, p. 306).

from them: what in Latin is more precisely called the *ratiocinium*.[13] It should also be noted that only a *human* intellect can properly be called "reason", because the term refers to the fact that the human intellect is inherently *discursive*: that is, it uses internal *words* (Greek: *lógoi*; Latin: *rationes* or *verbi*). The irrational animals do not have any intellect at all, and the angels (and God) have a strictly intuitive intellect that does not require discourse.

Now, as I will emphasize many times, logic studies neither the human intellect nor its operations (both of which would pertain to rational psychology or philosophical anthropology), but rather the *works produced* by that intellect, which, as we will see below are of three kinds: concepts, enunciations, and arguments. Nevertheless, since we are attempting to study logic in a scientific way, and the intellect is the *cause* of those works, it is important to understand what the intellect is and what it does.

In sum: faculties (as part of the larger genus of qualities that we call active potencies or active powers) are the first accidents to emanate from a substance. They provide the substance with the basic capacities to *do* things and in general to *act*. They are the *sine qua non* for any operation whatsoever. Since we are studying logic, which studies the works of the human intellect, naturally, the faculty that we are going to discuss the most is the intellect. We must now turn to the ways that faculties can, as it were, be "trained" (or deformed) in order to function better (or worse): that is, the moment has arrived to discuss *habitus* systematically.

Habitus

We are now in a position to talk about *habitus* more systematically, having touched on them in the previous chapter. In general, an *habitus* (Greek: ἕξις, *héxis*) can be thought of as a stable ordering of a substance, which either perfects it or deforms it.

Here is an analogy: suppose that I take a clean sheet of white paper. In itself, the paper has a large number of possible configurations: it could be flat, rolled into a tube, folded, or crumpled into a ball. Let's suppose I roll it into a tube, gently, without creasing it. If I wish to go back to a flat sheet, there is no particular obstacle: all I need to do is uncurl it.

Now, suppose that instead of rolling it, I *fold* the paper neatly into three parts, making two creases (the way one folds a letter). Now, if I ever wanted to go back to a flat sheet, I would have trouble, since the creases are hard to remove. On the other hand, the paper will never fit into a letter-sized envelope unless I first fold it.

This analogy helps us to understand what in philosophy we call *dispositions* (Latin: *dispositiones*; Greek: διαθέσεις, *diathéseis*) and *habitus*. When I roll up the paper gently,

[13] I will alert the reader from the outset that there is an ambiguity in the way that Aquinas uses the Latin term *intellectus*. In most cases, he means the faculty by which we know things as they are, but in a few cases he is referring to an *habitus* by which we know the first principles of reason. We will discuss that later on when we talk about the object of logic.

I do change something about it—I "perfect" the paper, as it were—but the tube-shaped configuration of the paper is transitory. Similarly, dispositions are transitory perfections of a substance that can easily be "undone." On the other hand, the creased fold causes a *permanent* change in the paper that is difficult to undo. Similarly, an *habitus* creates state in the substance that is difficult to remove.

Some examples are in order. Returning to the example we made earlier, all human beings, thanks to their rational nature, have the raw capacity to understand a science like quantum mechanics: that is, they all have the faculty that we call the intellect. However, as demonstrated by the fact that most people do not immediately understand the Schrödinger equation, our intellect needs "training" or formation in order to make accurate assertions about quantum physics. With some basic formation, we can acquire the correct *opinion* (*dóxa*) about certain facts—for instance, perhaps we have heard the idea that in the very microscopic scale, things behave both as waves and as particles—but this cannot be said to be a permanent and thorough assimilation. Armed with just that opinion, we will hardly be able to understand the two-slit experiment or the Pauli exclusion principle. In order to do that, one must be able (as we have seen) to trace the phenomena to their causes—which, in the case of quantum physics would entail deriving the relevant equations from the data and models that give rise to them. Only at that point is the assimilation secure and permanent—that is, truly a *scientia* (*episteme*), as we have understood the term. Right *opinion* about quantum physics is a disposition that enables the intellect to make *some* right assertions about the quantum world, in a transient and often unreliable way; acquiring the *science* of quantum mechanics, including tracing the phenomena to their causes, give secure, necessary, and permanent knowledge. Thus, a science is best described as an *intellectual habitus* that enables the one who has it to be secure in his knowledge of an object.

Notice that the intellect could also be malformed: I could gain *false* information about quantum physics, which could be transitory or deeply engrained. If the latter, it would consist in a kind of ignorance or prejudice: a deformation of the intellect. If this defect is durable and difficult to remove, it, too, is an intellectual *habitus*, but a bad one.

Thus, I have described four "perfections" (two of them better described as *im*perfections) of rational substances that affect their ability to use their intellect: two dispositions (true opinion, false opinion) and two *habitus* (science and deep-seated prejudice) that "train" (or malform) the intellect, enabling them to perform operations that they were unable to do before. When an *habitus* affects operation like this, we call it an *operative habitus* (*habitus operativus*); when that operative *habitus* is a good one (when it actually makes the substance more perfect) we call it a *virtue* (Latin: *virtus*, Greek: ἀρετή, *areté*); when it *deforms* that substance, rendering it less perfect, we call it a *vice* (Latin: *vitium*).[14]

[14]There is no special name for good or bad dispositions.

Perhaps you are more familiar with applying the terms *virtue* and *vice* to morality. We often say that a person is *virtuous* if he behaves in a way that is in conformity with the natural law. It would, however, be more precise to speak more specifically: a *moral virtue* is a good operative *habitus* that stably and durably helps a person to behave in a morally acceptable way, making such behavior easier to accomplish and even pleasurable. In contrast, a *moral vice* is a bad operative *habitus* that stably and durably pushes the person to commit immoral acts. Each of these types of *habitus* also has a corresponding *disposition*, which helps the person to behave in a morally good (or bad) manner—but in a transient and non-durable way.

Let's look at some concrete examples. Suppose that one day when I am at work, I find a $20 bill on my desk. I strongly suspect that it belongs to my co-worker. I could really use that money, but—after hemming and hawing for a while—I decide that really the morally good thing to do would be to return it to my co-worker. But if I am truthful, the decision was a close one; I nearly pocketed the money. You could say I performed a morally good act, and act of *honesty*, which is a species of *justice* (that is, of giving each one his due). However, I did so reluctantly. Now, suppose that my colleague has a tendency to lose money, often leaving it carelessly on my desk. Thus, at least once a day I have to decide whether or not to restore the money or pocket it. As it turns out, after a week, I have returned the money every time; and after a month, it becomes practically second nature to me. Soon, I actually experience pleasure in doing so, especially given my colleague's gratitude. At the end of this process, we can say that I have acquired the moral virtue of *honesty*: that is, I now have a stable and durable tendency to return lost money to its owner.

However, of course, my free human acts are contingent: if I wanted to, I could act *contrary* to my conscience. Let's go back to the first day, when I first found a $20 bill on by desk: suppose that, in this new timeline, my greed gets the better of me and I decide to pocket the money. I don't get caught, and my colleague never discovers where he has left the money. At this point, I have committed a morally unacceptable act: an act of *dis*honesty (indeed, a type of theft). However, I have not yet acquired a stable tendency to act this way. Unfortunately, over the course of time, I take advantage of my colleague's carelessness, pocketing hundreds of dollars over the course of two months. I get better at figuring out where he leaves the money, even scheming to get him to drop *more* money, and I take pleasure in my newfound "wealth." By now, I have acquired a stable disposition to theft, thus forming a *vice*.

Note that, in either case—with both the virtue and the vice—there is a period of transition during which it is still relatively easy for me to change course. In that intermediate stage, we can say that I have a *disposition* toward honesty (or dishonesty), but not yet a virtue (or a vice). Nevertheless, that disposition becomes stronger and more deeply seated as I continue to favor it with my action, until it "hardens" into an *habitus*.

These examples, I think, are sufficient to illustrate how operative *habitus* (and dispositions) work: in short, they are principles of operation that have the function of perfecting a faculty, giving it the ability to do things that it was unable to do before, or else to do them with greater ease. We can say that they are, as it were, halfway between the faculty and its operation: they partially actualize or determine the faculty, causing it to tend toward some operations over others.

Attentive readers will observe that up to now I have consistently specified that the *habitus* and dispositions we have talked about so far are all *operative*: that is, they are the principles of various kinds of *operation*. I have concentrated on the operative kind, because, of course, logic—our unique intellectual virtue that is simultaneously a speculative science, a practical science, and an art—is a intellectual operative *habitus*: evidently, it is a *good* one, hence, an intellectual virtue. However, I will mention in passing that there is also a whole class of dispositions and *habitus* that are not immediately ordered to operation. Rather, they produce a state of *being* in the substance. These *habitus* and dispositions are called *entitative*. To give a few examples, health and physical beauty are both entitative dispositions; and the sanctifying grace that Christians receive at Baptism is perhaps the best example of an entitative *habitus*.

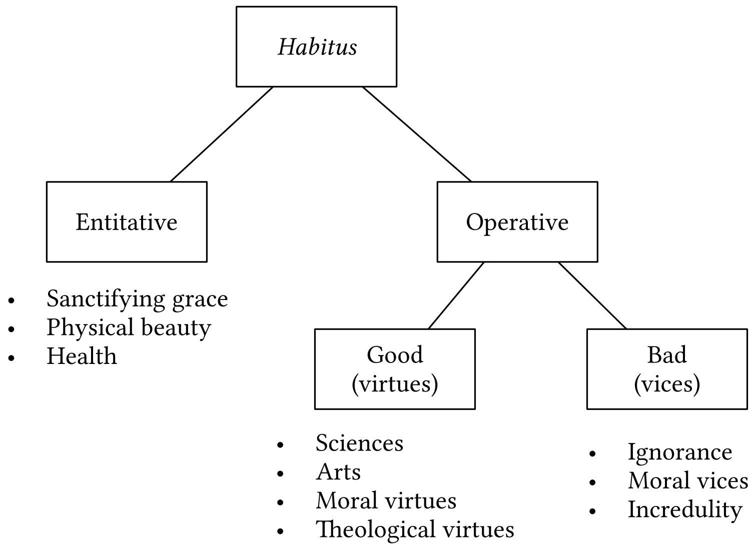

Figure 2.1: The divisions of *habitus* and some examples of each. Note that dispositions are divided the same way as *habitus*, and, in fact, physical beauty and health are best described as dispositions.

It is worth noting at this point some of the elements that operative *habitus*[15] all

[15]From now on, for brevity's sake I will speak only of *habitus*, because that is the genus that logic falls under. However, all of these characteristics apply equally well to dispositions, unless otherwise specified. As we saw, dispositions are like *habitus* in all respects, except for durability.

have in common, many of which they share in common with faculties. In particular, both operative *habitus* and faculties are inherent in a *subject* or substrate, and both are principles or originators of *operation*, and both have an *object*, that is, the finality or term to which the operation is directed.

As usual, it is helpful to look at examples. Let us look first of all at some faculties, starting with the faculty of sight. Like all faculties, the *subject* (or *substrate*) will be the substance in which it is inherent (in this case, an animal, rational or irrational). We can specify, however, that unlike purely immaterial faculties like the intellect, a sense like sight is dependent on the *body* or *matter* of the animal in question. The *operation* that sight produces is called *vision*, and the *object* of sight can be described as *colored material bodies insofar as they have color*. Now, let us look at the human intellect. Being also a faculty, it will likewise have the substance as its *subject* (in this case, the human being in question)—however, unlike the sense of sight, it stems directly from the human being's immaterial soul, with no dependence on the body for its existence. Its *operations* include the simple apprehension of essences, and making judgments (mediated or immediate) regarding being. (We will return to this point in greater detail.) Its *object* can be described as the entities it comes to know, insofar as they are intelligible (that is, insofar as they can be known in themselves).

With operative *habitus*, it is important to specify what we mean by *subject*: like all accidents, operative *habitus* inhere in the substance that they perfect, and so the substance could be regarded as their "subject" in that sense. However, more often, when we speak of the *subject* of an operative *habitus*, we mean the faculty that the *habitus* modifies. That is the meaning that we will give the term in this introduction. Thus, for example, all the intellectual *habitus* that we have talked about up to now—sciences and arts—have the intellect as their subject. If we look at, say, quantum mechanics, its *object* is what the science studies: material entities as regards the behavior of the elemental particles that constitute them, and the *operations* it produces consist in the acts of apprehending and judgment by the intellect regarding that object. Note that without the *habitus* of quantum physics, such apprehensions and judgments would be impossible; thus, the habitus is a true principle, or origin, of operation, albeit a secondary one that depends on the intellect and ultimately on the substance (that is, the human being) in which it inheres.

Similarly, if we look at an art—say, carpentry—the *subject* is, of course, the intellect; the *object* is the works of carpentry produced, which consists of pieces of wood to be shaped into tables, chairs, and the like; and the operations consist in the combination of knowhow and physical actions required to shape the wood into the products desired. Table 2.2 summarizes the examples we have just gone over, with some additions.

Table 2.2: The subject, operations, and object of various faculties and operative *habitus*.

	Subject	Operation	Object
Sight	Substance (body)	Vision	*Ens* as colored
Intellect	Substance (soul)	Apprehension, judgment	*Ens* as intelligible
Will	Substance (soul)	Desire, love	*Ens* as desirable
Quantum physics	Intellect	Apprehension, judgment (of Q.P.)	Material *ens* at subatomic scales
Carpentry	Intellect	Actions to shape wood	Wood to be shaped
Logic	Intellect	Apprehension, judgment, reasoning	Works of reason

Operations

Of course, the effect of having faculties and *habitus* is the capacity for *operation* (Greek: ἐνέργεια, *enérgeia*), which is simply the philosophical term for any actions that are produced by a substance. As we have seen, operations always have their *ultimate* origin in the substance itself, but the faculty that produces them, together with any *habitus* that perfects it, can rightly be considered the *immediate* origins of that operation. In our oft-cited example, the scientist who contemplates Schrödinger's equation is rightly considered the author or origin of the act (operation) of contemplating. However, he engages in this operation by the mediation of his intellect, which has been perfected by the science of quantum physics (an operative *habitus*).

Figure 2.2 illustrates the hierarchical relationship among substance, faculty, *habitus*, and operation. Operations can be produced entirely by the substance that acts, or else the substance may receive the power to act from an extrinsic source. For instance, an animal's capacity for locomotion is entirely intrinsic to itself; however, if it is trained to perform certain actions by a human master (say, for a dog to perform tricks or to sniff for contraband), it is acting based on training that it has received extrinsically. Similarly, a human being can do acts of knowledge and love under his own power; however, he can only do acts of supernatural knowledge (faith) and supernatural love (hope and charity) under the influence of grace, which he receives from God. However, logic deals with the works of reason, which are the result of operations that are entirely intrinsic to man and do not require an extrinsic power. As shown in the diagram, all operations that are proper—that is, intrinsic—to the substance that produces them ultimately have their origin in the substance's *actus essendi* or *act of being*. This is a principle that you will learn about in your course on metaphysics: but for now, it is sufficient to say that it is a metaphysical principle that is the source of all proper actuality (i.e., all the actuality that does not come from outside) in a substance. The *essence* (*essentia*) signifies the substance inasmuch as it is a metaphysical principle, a capacity that it fulfills in two ways: as the substrate for the accidents, and as the potency receives the act of being. As such, the essence acts as a limit to and a measure of the intensity of the act of being. The first perfections to emanate from the substance are the active potencies (*potentiae activae*), which, as we saw, include the faculties of sentient substances. Those faculties that entail a considerable freedom of action—especially the rational faculties in man—are further perfected by operative *habitus*.

Putting all this together, all operations emanate from the substance's *actus essendi*. Those operations that are entirely intrinsic trace all of their actuality to this act of being; others require the assistance of an outside power. Regardless, operations are produced by the mediation of substance's active powers. In sentient and rational beings, some of these powers—called *faculties*—are capable of a further perfection that we call operative *habitus*. Thus, a substance's operations pass through, as it were, two and possibly three filters: the essence itself (which already sets certain limits to what kinds of operations are possible), the faculties that produce the operations, and, where this is possible, the *habitus* that perfect the faculties.

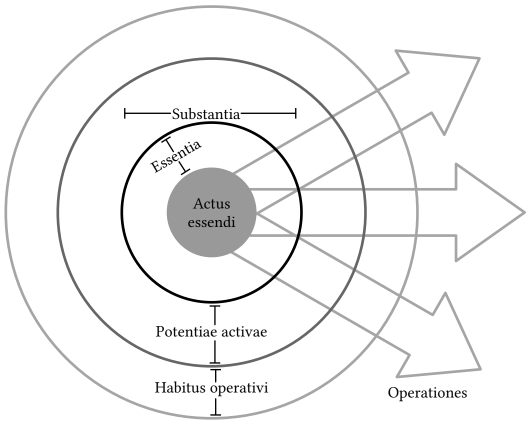

Figure 2.2: An illustration how a substance produces operations.

As a final note, every operation, evidently, produces an effect, which we call its *work* (*operatum*) or *product*. This work can be extrinsic to the substance that has produced it—for instance, the chairs and tables produced by the efforts of a carpenter—or else it can be immanent—such as the internal effects of the actions of our intellect and will. It is important to note (and I will reiterate this point later on) that *the works or products of an operation are distinct from the operation itself*. When it comes to logic, we will see that logic studies the *works* of the intellect, not the *operations* of the intellect, which would pertain to philosophical anthropology.

2.4 The Operations and Works of the Intellect

Even though the object of logic is the *works* of reason, it is still good for us to have an overview of what the intellect does. Fundamentally, the intellect (more specifically, what Aquinas calls the "possible intellect") is capable of accomplishing two kinds of operation: it can discover *what* something is, and it can discern *whether* something is the case. We say in more precise philosophical terms, that the intellect can discover a thing's *quiddity* (from the Latin *quid est*, "What is it?"), and it can make a judgment regarding that quiddity's *being*—that is, whether, in fact, that quiddity *is* or *is not*.

Simple Apprehension

The species of intellectual operation that discerns *what* something is, we term the *simple apprehension*. To illustrate how this operation works, we can do a simple exercise. Suppose I attempt to make the following statement:

> Ice cream.

This "statement" surely conveys some information to the reader; "ice cream" is a idea that we all understand. However, these words, by themselves, are insufficient to make an assessment regarding an actual state of affairs. You have no idea whether I am saying that I enjoy ice cream, or that chocolate ice cream is my favorite flavor, or that the ice cream that I have served you is melting. With the words "ice cream," I convey merely a *notion*, an answer to the question "What is it?"

In our discussion above, we have already seen something that answers the question "What is it?": namely, the substance, or more precisely, the *essence*—the ontological principle in a substance that gives it its definition. Through a process that you will see in your other courses, a faculty that Aquinas calls the agent intellect is able to *abstract* whatever is intelligible from the material entities that we encounter. This intelligible notion or definition (Latin: *ratio*) is what actually answers the question "What is it?" in our intellect, and thus we give it the technical name *quiddity* (Latin: *quidditas*). It should be noted that the quiddity is not a distinct "thing": it is not distinct from the object that is know. Rather, it is is merely that object *inasmuch as it is intelligible* to us.

Ontologically speaking, only the substance can strictly be an "essence"; that is, only a subsistent entity can answer the question "What is it?" and indeed, material substances are the objects that our intellects most readily apprehend. However, we are *also* able to understand other entities, such as the accidents of substances, and (through the use of negative judgments and analogy) even entities that transcend the material world, or that can only be understood by analogy. As a consequence, in *logic* (but not in metaphysics), it is necessary to expand our idea of "quiddity" to include notions that are not substances, and even some notions that are not completely unified—notions that are *analogical* rather than *univocal*, as we will see in class.

Thus, the simple apprehension is what enables us to understand *quiddities*, and the product or work of that operation we call the *concept*[16]. As we will see in class, the concept can also be called the *universal* (technically, the *universale in praedicando*), because a single concept allows us to know the quiddity of multiple individual things; and it can be called an *intention*[17], because it leads us toward the thing that is known. Ontologically, a concept is a "passion of the soul" (Latin: *passio animae*)—that is, the immanent effect on the soul of the act of apprehending. Since the concept is one of the objects of our study, we will study it in much greater detail in class.

Judgement

Clearly, the simple apprehension by itself leaves us with very incomplete knowledge: although it answers the question *what*, but itself, the apprehension tells us nothing about the actual state of affairs. Going back to our example, I can use the concept "ice cream" to make a series of propositions like the following:

- This (what I have in my hand) is ice cream.
- The ice cream is melting.
- The ice cream has a chocolate flavor.
- I ate ice cream yesterday.

Notice that in each statement, I am uniting two distinct concepts[18]—or as Aristotle would say, I am *composing* them. That is, I am affirming that the concept on the right-hand side (which we call the *predicate*) truly *is* a reality that pertains to the concept on the left-hand side (which we call the *subject*). Subject and predicate, in this case, are united, either explicitly (as in the first two statements) or implicitly (in the last two) by the verb *to be*, which we call the *copula*. By composing subject and predicate in this way, I am affirming the *being* of the predicate in the subject.

However, suppose that instead, I make the following statements:

- This (what I have in my hand) is not ice cream.

[16] From the Latin *conceptus*, that which has been conceived.

[17] From the Latin roots *in* (towards) and *tendo* (to stretch out, to proceed, to strive).

[18] Observant readers will note that although each proposition has a single subject and a single predicate, some of the predicates are, in fact, *compound*—that is, they unite two concepts into one. For example, if I were to draw out the last proposition into its elementary enunciations, they would be something like: "I ate; the eating took place yesterday; ice cream was eaten." Our language makes shortcuts that greatly simplify the expression of our judgments.

- The ice cream is not melting.
- The ice cream does not have a chocolate flavor.
- I did not eat ice cream yesterday.

As before, I have taken two concepts, which can be termed *subject* and *predicate*; however, this time, I have *denied* the being of the predicate. Logically speaking, instead of composing subject and predicate, I have *divided* or *separated* them.

From the discussion above, I think it is clear why Aristotle and Aquinas call this operation *composition and division*, because it always takes two concepts—a subject and a predicate—and either unites or separates them. It can also be termed the "judgment" (and this was the terminology preferred in the Modern period). Like the simple apprehension, the judgment produces a work that is immanent to the soul, which we call an *enunciation*. Just as concepts are represented by terms, enunciations are represented verbally as *propositions*, like the statements I made above. It should be clear that it is only after making a judgment and producing an enunciation that it makes sense to talk about truth and falsehood. Those enunciations are true that correspond to reality, and likewise, those that are in disagreement with reality are false. That is, if I affirm something that actually exists, then my affirmation is true; similarly, an enunciation denying something that does not exist is also true. However, if I affirm something nonexistent, or deny something that does exist, my enunciation is false. Once again, it is the *product* or *work*, immanent in the soul, that is the object of study of logic.

Reasoning

When we make a judgment regarding being, we can arrive at it in two ways: either immediately, simply by seeing it, or by the mediation of judgments that we have previously made. For example, any judgment that I make regarding things that I can verify directly with my senses—say, the fact that the pine tree outside my window is green—is immediate. The intellect, by acts of apprehension, generates the concepts "pine tree" and "green," and by means of a judgment constructs a relation of attribution between them, thus producing an enunciation: "The pine tree is green." In other words, producing that enunciation does not require anything except the apprehension of those concepts and verifying that they do, in fact, exist in reality. Suppose, however, that having learned about the biology of pine trees, I make the following enunciation: "Pine trees are green because the chloroplasts in the pine needles predominantly absorb all of the wavelengths found in visible light *except* green."[19] There is no direct way to verify the existence of chloroplasts, still less the fact that they tend to absorb red and blue light more than green light. Reaching this conclusion requires highly specialized

[19]Attentive readers will note that this enunciation, unlike the first one, is actually a *compound* enunciation that includes a number of simpler ones joined together.

instruments and a solid foundation in cell biology. In short, it is impossible to make a judgment regarding the being of the chloroplasts without extensive use of *previously established enunciations*.

The discursive process that we use to make judgments that rely on previously obtained knowledge is called *reasoning* or *argumentation* (Latin: *ratiocinium*). As we will see in the course, argumentation entails constructing a series of *arguments*, which consist of two previously known enunciations called *premises* that are linked by a concept in common called the *middle term*. By linking the two enunciations and dropping the middle term, we produce a third enunciation called the *conclusion*. As with concepts and enunciations, arguments are internal, intelligible passions of the soul that are distinct from their external, verbal expressions, which we call *syllogisms*. For example,

All *men* are mortal.
John is a *man*.
Therefore, John is mortal.

Of course reasoning made in real life will involve a great many arguments, but every reasoning can be broken down to simple arguments like this one, which, in its verbal expression, is called a *categorical syllogism*. Note that the intellect is not employing a third operation, exactly; it is still using the second operation, the judgment. What changes is merely the means that it uses to arrive at its judgment on being: that is, previous knowledge, as opposed to direct evidence.

Table 2.3: A summary of the operations of the intellect, their work, and what they enable us to understand.

Operation		Work	What It Understands
Simple apprehension		Concept	*What* something is (quiddity)
Judgment	immediate	Enunciation	*Whether* something is (being)
	mediate	Argument	

Table 2.3 summarizes the role of the two operations of the intellect and the three works they produce.[20] These works—concepts, enunciations, and arguments—form the basis of the three branches of logic: logic of the concept, logic of the enunciation, and logic of the argument (or syllogism). At this point, it bears reiterating: *logic studies*

[20] According to some logic textbooks, there are *three* operations of the intellect: apprehension, judgment, and reasoning. In my opinion, this is not entirely correct. It is better to say that there are two operations: apprehension and judgment. However, in order to arrive at some judgments, arguments must be employed.

the works of reason, not its operations. What we will be discussing together are concepts, enunciations, and arguments; it is not our task, in this course, to consider apprehension and judgment, except insofar as these are the *cause* of the works. In fact, these works form the *object*—or, as we will learn to call it, the *material object* of logic. We will discuss the object of logic in detail in the next chapter.

Chapter 3

The Definition of Logic

In this chapter, we will come to a detailed understanding of the definition of logic, and thus we will finally be in a position to answer the question *What is logic?* definitively. I will go ahead and state the definition here, but, of course, we will explain each part of it in detail:

> Logic is the instrumentally speculative and modally practical science whose material object is the works of reason, whose formal object *quod* is the relations of reason constructed between second-order intentions, and whose formal object *quo* is the light of the first principles of reason.

We have already laid the necessary groundwork. In fact, you should already be able to explain what it means for logic to be an *instrumentally speculative and modally practical*. By the end of this chapter, you should have a good understanding of what logic actually is.

3.1 The Object

As we saw above, all faculties and operative *habitus* have an *object*; that is, the entities that are referred to that faculty or *habitus*' operation. We gave a number of examples above, but we need to look at the notion of object a bit more carefully. Let us take, for example, the faculty of vision, which we said has as its object colored bodies inasmuch as they are colored, or as Aquinas puts it,

> Now, properly speaking, we call something the *object* of a faculty or *habitus*, when all things are referred to that faculty or *habitus* by reason of that thing. For instance, *man* and *stone* refer to sight insofar as they are *colored*. Therefore, *colored things* are the proper object of sight.[1]

[1] *S.Th.*, I q. 1, a. 7, *responsum* (my translation).

Notice how the object practically defines the faculty: vision is the sense that allows us to see colored things. Let us unpack this a little, because it will help us to understand how to describe the object of an art or science like logic.

We said in the first place that the object of vision is colored material bodies, or *res colorata*. These bodies constitute, if you will, the set of all things that can be the object of sight: it is, obviously, impossible to see anything that is immaterial or invisible. When a material entity becomes the object of our sight, it does so thanks to a quality called *color*: that is, an active power that allows the object to emit a medium that our eyes can detect and that our brain can interpret. Finally, there is the means that is used to generate color, which is also the medium that transmits it to eyes: that is, *light*.[2]

This brief analysis suggests that we may discern three aspects of the object of sight: *res colorata* constitutes what we will call the *material object* of the sense of sight; *color*, which is the aspect of colored bodies that our eyes are able to detect, we will call the *formal object quod*;[3] and light, which is the means by which we see the object, we will call the *formal object quo*.[4]

Let us unpack this further. As we saw in the last chapter, in metaphysics *matter* refers to a metaphysical principle that, by itself, lacks definition or actualization, and requires an active principle, a *form*, in order to be something. We saw, for instance, how substances function as a sort of secondary matter in which the accidents are inherent (as when marble takes on the figure of David or Apollo); and also how the human person is composed of a *prime matter*, the body, that is informed by a *substantial form* that we call the *soul*.

When we use the terms *material object* and *formal object*, we are, of course, applying the metaphysical concepts of matter and form by analogy. The material object is "material" in the sense that, by itself, it lacks definition: it is not "material" because colored bodies contain the ontological principle called prime matter (which, as it happens, they do), but because it could not be presented to our sense of sight without the particular aspect that *makes* it apt for vision. As we saw, that specific of *res colorata*

[2]The ancients did not know that light is a kind of electromagnetic radiation, with a wavelength between that of infrared and ultraviolet radiation, that is, between 400 to 700 nanometers. As it turns out, unless you are viewing light emitted directly by a light source, the color of an object is produced when light from a light source (usually one that emits white light with a wide range of wavelengths) is absorbed by that object and is partly reflected into your eyes. The object tends to absorb some wavelengths better than others and reflect the rest. Thus, if the object is red, it is predominantly absorbing light with broadly blue and green hues, while reflecting back mostly red hues. Our eyes, in turn, fundamentally have three types of color detectors called *cones*. Each type is most adept at detecting light of a particular hue: red, green or blue. By combining the signal detected by each kind of cone, the brain is able to construct the full range of colors that we are familiar with. None of this fundamentally changes the fact that there is a *quality* of the object that is seen that is able to receive the incident light, that under the action of the light becomes an active power that then transmits the emitted light to our eyes. We call that quality *color*.

[3]In Latin, the *obiectum formale quod*; literally, the "formal object *that*" we see.

[4]In Latin, the *obiectum formale quo*; literally, the "formal objet *by which*" we see.

is precisely *color*. Because it gives the material object its definition, we call color a *formal* object, and we call it a formal object *quod* because color is *that which* our sight actually sees. We can go even further and specify the *means* by which we see color—namely, light—and since this functions as a kind of efficient cause of our seeing, it is, as it were, even more "formal" or determinative than the formal object *quod*. For this reason we call light the formal object *quo*, because it is the means *by which* we see.

Let us look at another example that is, perhaps, closer to our goal: an art like carpentry. As we saw previously, every art is a kind of intellectual *habitus* that permits the artist to perform certain operations (not necessarily strictly intellectual in nature) that he would be unable to do without possessing the art. In this case, the set of objects that we are interested are the things that the carpenter will actually work with; that is, things made of *wood*. Thus, we can say that the material object of carpentry is *wood*. Of course, the purpose of carpentry is to transform raw wood, as it is obtained in the lumber yard, into finished products such as tables and chairs. Thus, we could say that the formal object *quod* of carpentry is wood *insofar as it is capable of being shaped*. Poor quality wood, or wood that has already been made into a finished product, is most likely *not* something that the carpenter will use as his raw material (unless perhaps he is fixing or restoring it). Finally, the formal object *quo* consists of the various techniques that the carpenter uses in order to make his finished products.

When the *habitus* in question is a science, the material object will be what Aquinas calls the *genus subiectum* and Aristotle τὸ γένος τὸ ὑποκείμενον (*tò génos tò hypokeímenon*): the *subject matter*, or the set of things that the science studies. Thus, for example, biology studies *living things*, and these constitute its material object. The formal object *quod* of a science will be the particular aspect of that material object that is of interest to the science. For instance, biology does not study living things in every respect—it does not, for example, look in to the metaphysical principles of living things—but rather only it studies them *insofar as they are alive*. Thus we could say that the formal object *quod* of biology is the *vegetative and animal life* or *bodily life* of the living things it studies. Finally, the formal object *quo* of a science will be the method used: in the case of biology, the scientific method, which entails observation, forming hypotheses, making deductions, and conducting experiments and other date-gather activities. Table 3.1 summarizes these examples.

You will often see a science described using a formula such as the following: "Science S studies entity E inasmuch as it has property X." For example, "Metaphysics is the study of entities (*ens*) inasmuch as they *are* (*qua ens*)," or, "Biology is the study of living things inasmuch as they have bodily life." This is simply a short way to define a science in terms of its material object (entity E) and formal object *quod* ("inasmuch as it has property X").

One final observation: there is *one* object of a given faculty or *habitus*. The language that we have used here might give the impression that there are three objects, but in reality these are three *aspects* of a single object: the things that we have identified as

Table 3.1: Examples of the objects of various faculties and *habitus*. Note that I have listed some faculties first, followed by arts, and finally some sciences.

	Material Object	Formal Object *Quod*	Formal Object *Quo*
Sight	Colored *Ens*	Color	Light
Intellect	*Ens*	Intelligibility	The first principles of reason
Will	*Ens*	Desirability	The first principles of morality
Carpentry	Workable wood	Suitability for being shaped	Woodworking techniques
Painting	Things that can be depicted	Ability to be depicted	Painting techniques
Biology	Living things	Material life	Scientific method
Quantum physics	Material *ens*	Subatomic scale	Scientific method
Metaphysics	*Ens*	*Ens*	Resolutive reasoning

referred to the faculty or *habitus*. Thus, to reiterate, these things are *materially speaking* some sort of entity (real or in our reason alone, as we will see) and we call them the *material object*. Formally speaking there is some specific *aspect* that refers these entities to the faculty or *habitus*, which we call the *formal object quod*. Finally, there is a means that permits the referral to take place, which we call the *formal object quo*.

3.2 The Material Object of Logic

We must now, of course, turn our attention to logic itself. What is the material object of logic? The answer, fortunately is relatively straightforward. Logic studies the *works of reason*, which I have outlined in the previous chapter: concepts, enunciations, and arguments. These, as we saw, are *passiones animae*, that is, qualities of the soul that remain immanent there, due to the operations of the intellect (which, as we saw are the *apprehension* and the *judgment*, the latter operating either immediately or mediately).

I will take a moment to emphasize once again: *logic studies the works of reason, not the operations of reason*. We are interested in forming precise and well defined *concepts*, *enunciations* that are properly formed and expressed, and *arguments* that are both correctly constructed and true in their conclusions. In logic, the manner in which the intellect functions does not enter the object as such: we are only interested in the intellect's operations as the *causes* of that object.

We must, however, now tackle the much more difficult issue of the *formal object quod* of logic: what is it about the works of reason that logic concerns itself with? What *aspect* of these works does it focus on?

3.3 The Formal Object *Quod* of Logic

As the definition I gave above states, the formal object *quod* of logic regards *relations of reason* that we construct between second-order intentions. Before we get any further, we will need to discuss *relations*, *relations of reason*, and *intentions*.

Relations

Consider for a moment a father and a son. Metaphysically speaking, both are, of course, substances. However, once the father has begotten a son, he acquires a new perfection that he did not have before: *fatherhood*. Likewise, the son simultaneously receives a perfection upon his conception called *filiation* or *sonship*. Those perfections are not, themselves, substances, but they are real attributes; hence, they must be what we have learned to call *accidents* in the last chapter. In fact, Aristotle considers perfections such as these to be their own category, or supreme genus, of accident that he calls the πρός τι or "toward which."

In short a πρός τι, or, as we have come to know it through Medieval Latin, a *relation* is, in the first instance, a reference that one substance has to another.[5] You will learn more about relations in your logic course, since they constitute one of the categories that we will study in detail. As we are about to find out, the concept of *relation* is extended beyond that of the real perfection that is inherent in a substance. We sometimes call this kind of relation a *real relation* so as to distinguish it from others that we are about to look at. Table 3.2 gives some examples of real relations.

Table 3.2: Some examples of real relations. Note that relations are always conceived *bilaterally*: that is, the relation from the subject to the object will normally be coupled to a relation from the object to the subject. However, in some cases, the reciprocal relation is not a real one. The relation "having known" is real; the reciprocal relation, "being known," is not.

Relation	Subject	Object	Foundation
Fatherhood	the father	the son	generation
Sonship	the son	the father	generation
Double	a two-meter-long substance	a one-meter-long substance	quantity
Half	a one-meter-long substance	a two-meter-long substance	quantity
Having known	the knower	the think known	intentional similarity

[5] For anyone who has studied Latin, it can be helpful to recall that the word *relation* comes from the verb *re-ferro*, which takes the past participle *re-latum*.

As I said, we will see this in detail in the course. For now, a few observations about relations are all that we need. First, some terminology: the substance in which the relation is inherent we call the *subject* of the relation (as we would for any accident). The substance referred to we call the *object*. Moreover, the relation will always be based on some kind of *foundation* (quantity, action and passion, or some kind of knowledge). Second, real relations are normally *bilateral* or *reciprocal*. That is, the relation inherent in a subject will ordinarily be paired with *another* relation that is inherent in the substance that is the object of the first relation. The example we gave above of fatherhood and sonship illustrates this well: a father cannot be a father unless he has a child (in our example, a son), and that son must necessarily have a father. For this reason, relations are always *conceived* as bilateral "by default" by our intellect—even when it turns out that the reciprocal relation does not really exist. (We will see some examples below.) Finally, in modern parlance, we tend to refer to the *pair* of relations as a *relationship*. For Aristotle and Aquinas, however, what we call a *relationship* actually consists in two distinct *relations*, each one a perfection inherent in a distinct subject. Thus, for example, in modern parlance we would say that a husband and wife have an indissoluble *relationship* that we call the *marriage bond*; however, in metaphysical terms, that bond consists in *two* relations, one inherent in the husband, and the other inherent in the wife. A similar things can be said, say, of the father–son relationship and the teacher–student relationship.

Relations of Reason

I have brought up real relations at this point not because they are directly pertinent to our topic, but because they are necessary for understanding an essential element of the definition we are considering: *relations of reason*.

A *relation of reason* (*relatio rationis*) can be though of as a construct that our reason treats *as if* it were a real relation, when in reality it is nothing at all. A few examples will illustrate the idea. Consider the following proposition:

> John is himself.

What we have done here is subtle. The subject of the proposition is *John*, but the predicate is also *John*: the term *itself* is merely a placeholder to avoid being repetitive. Thus, *John* is, in a sense "divided" by our reason; that is, we consider two "instances" of him and attribute to each one a kind of relation. In effect we are saying that John (in the subject) is *the same* as John (in the predicate), and vice versa. However, this relation, which we can call *sameness* or *identity*[6], does not actually exist. John (in the subject) is not really distinct or different from John (in the predicate). The relation

[6] From the Latin *idem*, meaning "the same". Readers preparing to write papers should be familiar with the Latin term *ibidem* (from *ibi*, "there", and *idem*), which means "in the same place."

of *identity*, therefore, does not really exist, but rather is merely a *construct* to help us understand that substances are united and permanent—that is, a substance (or for that matter, any entity) cannot simply become something else without some kind of real transformation. *Identity*, therefore is a *relation of reason*, not a real relation.

Or else, consider real relations that are based on knowledge. Suppose, for example, that you have read a book—say an introductory textbook on logic. Now that you have read it, the relation of *having read* the book is now inherent in you. However, the book, as such, is completely unchanged by you having read it. As I mentioned earlier, we are unable to conceive of a relation in any other but as bilateral or reciprocal, and so our reason cannot help but consider the relation (ostensibly inherent in the book) of *having been read* alongside the real relation. In other words, when you read a book, you acquire a new perfection, *having read* the book; the book acquires nothing at all, but we cannot help but conceive a *having been read* in tandem with *having read*. However, we use our judgment to deny that *having been read* actually exists: it is, in short, a *relation of reason*.

In order to understand these relations of reason, it helpful to mention that they belong to a broader group of pure objects, or constructs, that we call *entia rationis*, entities of reason. An *ens rationis* refers to any construct that we treat *as if* it were an entity, when in reality it is not. It is, if you will, a placeholder concept which, upon examination, does not refer to any actual *thing* (*res*) in reality.

As usual, it is easiest to understand the idea using examples. Consider *hobbits*. If anyone is not familiar with the idea, hobbits are a fictional people imagined by J.R.R. Tolkien in his fantasy works, *The Hobbit* and *The Lord of the Rings*. In reality there *are* no hobbits. They are not, in the strict sense *essences* or *substances*; however through our imagination, we can describe what such a people would be like if they *did* exist. In other words, hobbits are *figments*, with no actual ontological foundation. In fact, this analysis is valid for any fictional character or object: Greek gods, superheroes, Romeo and Juliet, light sabres, and so on.

On the other hand consider a construct such as *darkness*. Darkness is not really an entity; it is best characterized as the *lack* of light. It is, however, clearly based on real situations: darkness does not exist as such, but dark *rooms*, dark *nights*, and dark *clothing* do. A similar thing can be said of other privations: *coldness*, *evil*, *blindness*, and the like. In each of these cases, we are, as it were, hypostatizing the *lack* of a real thing: *coldness* is the lack of heat; *evil* is the privation of good;[7] *blindness* is the privation of sight in something that ought to be able to see. There are also the relations of reason that I mentioned above, such as *identity* and *having been read*.

A common thread among all the *entia rationis* in this second group is that although

[7] It might not be immediately obvious that evil is a privation, but if you think about it, there are only evil *persons*, evil *circumstances*, and evil *actions*—that is, real entities that have somehow been corrupted. However, evil, in and of itself, does not exist.

each one is a pure object that, as such, does not exist, it nevertheless has a very precise *foundation* in reality, one that is entirely lacking in the first group. These examples suggest a twofold division of *ens rationis*: one according to the how the *ens rationis* is attributed to a subject, and the the other according to the way the *ens rationis* is constituted as a pure object of reason.

Let us unpack this idea. When I use *hobbit* as a predicate—as in the proposition "Frodo Baggins is a hobbit," the subject to which I am attributing the concept *hobbit* is completely nonexistent. There is, and never was, a hobbit named Frodo Baggins. On the other hand, when I say, "The room is dark," the subject, *room* is quite real—even though the attribute, as such, is not. We can thus divide *entia rationis* into those that are attributed to completely nonexistent subjects, and those whose subjects are real. We call former class *ens rationis sine fundamento in re* (the entity of reason without any foundation in reality), and the latter *ens rationis cum fundamento in re* (the entity of reason having some foundation in reality).

If we look only at *entia rationis cum fundamento in re*, our examples suggest that there are two ways to form them: either positively (that is, based on an affirmation), or negatively (based on a negation). It is not hard to see, based on the examples I mentioned, that the negation of some real perfection, as such, is not truly a "something," but rather the *lack* of something. As regards those *entia rationis* formed by an affirmation, Aquinas argues that when we affirm something in an absolute sense, that affirmation must refer to something in reality: as when we say, "The sky is blue," or "Gold has a density of 19.3 g/cm^3."[8] The only way to obtain an *ens rationis* through positive affirmation is to affirm it *relatively*—that is, as a relation of reason, as in the

[8]There is a difficulty in interpreting the passage that talks about this distinction, which is in *Quaestiones Disputatae de Veritate*: "Now, what exists in reason alone [*id quod est rationis tantum*, that is, *ens rationis*] can only be of two kinds: that is, either a negation or a kind of relation. For every absolute affirmation signifies something found in the nature of things [i.e., reality]. Thus to *ens*, 'oneness' adds something that is in reason alone, that is, a negation: an *ens* is said to be 'one' insofar as it is undivided. However 'truth' and 'goodness' are attributed positively; therefore, they cannot add anything but a relation that is in reason alone" (Thomas Aquinas, *Quaestiones disputate de Veritate*, Leon. vol. 22, Sanctae Sabinae, Rome 1975–1970–1972–1973–1976 [henceforth cited as *QdV*] 21, 1, *corpus*; my translation).

In this passage, Aquinas is talking about the *trascendentia* or the "properties" of *ens*, which later philosophers call the *transcendentals*. We will study these in the logic of the concept, but the relevant connection to *ens rationis* is that the notions conceptually distinguishing, say, oneness, truth, and goodness from *ens* are *entia rationis*—that is, they are not real things *in rerum natura* (as Aquinas would say). However, those distinguishing notions do have a real foundation: they are attributed to real things, namely, all *entia*. Aquinas' argument poses a difficulty: does he intend divide all *entia rationis* without exception into negations and relations? If so, what are we to do with *entia rationis sine fundamento in re*—those *entia rationis* with no foundation in reality, such as hobbits and Greek deities?

There are fundamentally two possible solutions: either *entia rationis sine fundamento in re* can in some way conceived as relations of reason, or else he is not considering them in this division. I tend to favor the latter interpretation: in short, I think that our reason treats *entia rationis sine fundamento in* example of the relations of *identity* and *having been read* that I mentioned above. Table 3.3 summarizes the divisions of *ens rationis*.

Table 3.3: A summary of the divisions of *ens rationis*.

	Division by attribution	Division by object	Example
Ens rationis	sine fundamento in re		hobbits
	cum fundamento in re	negation	blindness
		relation	having been read

Before moving on, I will make one final note about the term *ens rationis*. I focused on the meaning used most often in philosophy—that is, a nonexistent construct that our reason treats *as if* it were a thing in reality—however the term itself is ambiguous. The term "entity of reason," according to its verbal construction, simply means an entity that depends on reason in some way. Thus, in theory, it could have three possible meanings:

1. It could refer to the *real* entities that have have reason as their *efficient cause*: that is, it could be a synonym for the works of reason, namely, concepts, enunciations, and arguments. In that case, we say that their dependence on reason is *effective*.

2. It could refer to the *real* entities that have reason as their subject or substrate: that is, all of the intellectual *habitus*, such as sciences, arts, prudence, and the theological virtue of faith. In that case, we say that their dependence on reason is *subjective*.

3. It could refer to the *nonexistent* entities—in reality, pure objects or constructs—that our reason apprehends *as if* they were entities. In this case we say that their dependence on reason is *objective*.

It goes without saying that all of the *entia rationis* that we have talked about in this section are of the third kind, those that depend on reason *objectively*—that is as pure objects or constructs with no ontological consistency whatsoever.

With all of this background out of the way, the question remains: what are the relations of reason that interest us in logic? In essence, they are the relations that our reason constructs from one work of reason to another: from a concept to a concept,

re in a hypothetical way, *as if* they were posited absolutely—and then, with the judgment, denies that they actually exist. A reader of *The Hobbit* asks himself, "What would Middle Earth have been like *if* it existed and a hobbit named Bilbo Baggins lived there?" Thus, I think such complete figments are excluded from the division that Aquinas is considering here.

and from an enunciation to an enunciation. Here are a few examples that you will study during the course:

- Attribution or predication: the relation by which we affirm (or deny) the being of the predicate in the subject of an enunciation.
- Being the subject to which a predicate is attributed.
- Predicability: the five ways in which a predicate can be attributed to a subject.
- The relation of a species to the genus that contains it, and vice versa.
- The relations of the major and minor premise to the conclusion, and vice versa, which constitute the logical nexus.

Since all of these relations have works of reason (which are accidents, *passiones animae*) as their subject and object, they are clearly not *real* relations, but relations of reason. These relations will be the focus of our study throughout the course, and identifying or constructing them *correctly* will be our goal.

Intentions

When you hear the word *intention*, you are probably used to the most common modern usage of the term, which references the movement of the will: one *intends* to go shopping if one has *plans* to go to the store later on, we apologize for hurting someone *unintentionally* (i.e., without willing it), we make prayer *intentions* (i.e., express our intent to pray for someone), and so on. As I mentioned, in all of these cases there is a movement of the will toward some object. *Intention*, as you can probably tell, comes from the verb *intendō*, from the roots *in* (in this case meaning, "toward") and *tendō*, meaning "to stretch, proceed, or strive for." Among several meanings, *intendō* refers to a movement toward something: to turn one's attention to something, or (as in modern language) to plan to do something.

In Aquinas, *intendō*, and the related term *intentio*, can refer to movements of the will, but it can *also* refer to the operations of the intellect. In particular, it can refer to the *concept* insofar as it is directed to an *object*. For example, the concept "man" represents all human beings, and in that sense is directed toward (*intendit*) them.[9] For this reason, Aquinas uses the term *intentio* sometimes as a synonym for the concept, not just as referring to the movement of the will.[10]

Now, it is possible to form a concept (that is, an intention) of *real entities* (which is what we normally do), but it is also possible to form a concept of *the works of our reason*.

[9]Attentive readers might realize that there is, in fact, a *relation of reason* that refers the concept to its object. We call it "intentional similarity." The *real* relation inherent in the person who knows something through this concept we call *intentionality*.

[10]In fact, *intentio* in Aquinas can refer to the direction of anything toward its end: it is, in this sense, the "intention" (in modern language, "tendency") of the well aimed arrow to reach its target. See, for instance, I-II q. 12, a. 2, resp.: "intentio, sicut ipsum nomen sonat, significat in aliquid tendere."

For example, when I make the statement, "All human beings are mortal," the term *human being* there is referring to real flesh-and-blood human beings. On the other hand, I can also look reflexively on the works of reason that permitted me to make the statement: in this case, the concepts *human being* and *mortal*, and the enunciation that composed the two concepts together.

We call concepts that refer *immediately* to real entities *first-order intentions*. On the other hand, concepts that refer to some work of our reason, we call *second-order intentions*. Note that even second-order intentions refer to reality in a *mediate* or indirect way: for instance, when I take the term "human being" to mean the *concept* formed in my reason, that concept refers in the first place to the *concept*—but of course that concept, in turn, refers to real human beings. I will let Aquinas speak here:

> Something in reality can correspond to a concept in two ways. In the first way, it does so immediately, for instance when the intellect conceives the form of some thing that is found outside the soul, such as a man or a stone. In the other way, it does so mediately, for instance when something follows the action of the intellect, and the intellect reflecting upon itself considers it. In that case, the reality corresponds to that consideration of the intellect in a mediate way, that is, by means of the understanding of reality. For example, the intellect understands the nature of animals in a man, in a horse, and in many other species. From this, it follows that the intellect understands that nature as a genus. Nothing in external reality immediately corresponds to the concept by which the intellect understands the genus. However, some reality does correspond to the understanding from which this intention flows.[11]

Perhaps an analogy that can help is to compare the intellect with an instrument like a telescope. Just as the telescope is the instrument by which we are able to see faraway objects like stars, the intellect is the instrument by which we are able to know things. In order for us to see the stars, the telescope must focus the light using lenses and similar equipment. In a way, our concepts play a role analogous to that of the lenses: they are, if you will, the "window" through which we see the realities that are outside our soul. So long as the lens is focusing light coming from *stars*, it is similar to a *first-order intention*: that is, it is helping us to see, in an immediate way, the *stars*—an external reality. At this point, we will need to bend the analogy a little. Suppose that we could set up the telescope in such a way that we could mount a lens capable of focusing on *another lens*—perhaps in order to observe the first lens in action. That is how a second-order intention works: it reflexively looks back on the intellect, and points to *another work of reason* rather than an reality outside the soul.

[11]THOMAS AQUINAS, *Quaestiones disputate de potentia*, in: *Quaestiones disputatae*, vol. 2, Marietti, Turin 1965 [henceforth citied as *QdP*] q. 1, a. 1 ad 10 (my translation).

As usual, the distinction between first-order and second-order intention will be much easier to understand with examples. Consider the following two propositions:

- All men are mortal
- "Man" is a species of the genus "animal."

In the first proposition, the concept *man* is a first-order intention, because it refers to flesh-and-blood men. In the second proposition, *man* is a second-order intention, because it signifies, not flesh-and-blood men, but the *concept itself*.

In summary: a first-order intention is concept that refers directly to a reality outside the soul. A second-order intention is an intention that refers to another intention (or, at any rate, to some work of reason).

Relations of Reason between Second-order Intentions

We are finally in a position to understand the formal object *quod* of logic:

> the relations of reason constructed between second-order intentions

In short, *logic in itself does not study realities that are outside the soul*. Instead, it *studies the works of reason as such*, and as a consequence, we will be employing second-order intentions. Logic does not concern itself with the density of gold—that is for chemistry. However, it does concern itself with the fact that in the proposition, "Gold has a density of 19.3 g/cm^3," the concept *gold* is a species, it functions as the subject of the enunciation, and a predicate (density of a certain magnitude) is attributed to it as a property. As the example demonstrates, logic deals specifically with the relations of reason that the intellect constructs from one second-order intention to another (e.g., the relation of reason that makes *gold* the subject with respect to *density of 19.3 g/cm^3*, and the reciprocal relation that makes *density of 19.3 g/cm^3* the predicate of *gold*).

What logical relations will we study? They are fundamentally of three kinds or genera, which correspond to the three types of works of reason:

1. *Predicability*, which describes *concepts*, and concerns the ways in which a concept can be attributed to its subject. As we will see, concepts can be either *univocal* (with one well defined meaning) or *analogical* (having only a partial unity). Moreover, the univocal concept is predicable on one of five ways: as *species*, *genus*, *difference*, *property*, or *accident*.

2. *Predication*, which concerns actual attribution of a predicate to a subject, thus constituting an *enunciation*. We will learn to divide enunciations according to their mode of attribution by *quantity*, *quality*, *unity*, and *matter*.

3. *Illation*,[12] which concerns the logical consequence of a conclusion from its premises, thus constituting an *argument*. Illation can, in turn, be *deductive* (proceeding from general premises to more specific conclusions) or *inductive* (proceeding from specific premises to more general conclusions). We shall spend the majority of our study of the logic of argumentation on deductive reasoning, since this forms the backbone of all scientific argumentation.[13]

3.4 The Formal Object *Quo* of Logic

We have discussed the material object and formal object *quod* of logic: that is, *what* logic studies—namely, the works of reason—and *what specific aspect* of the material object it is concerned with—namely, the second-intention relations of reason that allow us to know the truth with certainty. We must now discuss *how* logic is able give us the ability to construct these relations correctly and thus bring us unfailingly to the truth.

As you will learn in your epistemology class, the first concept that the intellect makes is that of *ens*: that is as soon as the intellect is active, it realizes that there is *something* outside itself, and that this something *is*. We do not start out understanding *ens* in a formal or explicit sense: systematic study of metaphysics is required for that.[14] Instead, this knowledge is implicit, acquired by the very fact that our intellect is working: *actu exercito* (through the act that is exercised), as Medieval philosophers would put it.[15]

As soon as we acquire our first concept, *ens*, our agent intellect causes us to form our first enunciation at the very same moment: namely that *ens* is not *non-ens*, in other words, whatever it is that is presenting itself to the intellect must, on some level, *be*, and cannot possibly *not be*, since it is presenting itself right now. If, for example, I were to get up in the middle of the night and stub my toe on some hard object, the first thing I would know is that my toe has struck *something*, some entity, even if I do not immediately know what it is. It is also clearly not *nothing*: that object, which is

[12] Note that "illation" comes from the Latin verb *infero* (past participle: *illatum*), from which the word *inference* is derived.

[13] As we will learn, we cannot arrive at certain, universal knowledge by relying solely on singular data. Pure induction, therefore, only produces *probable* conclusions. However, contact with even one material entity gives some knowledge about that thing's *nature* or *quiddity*. That knowledge is sufficient to make universal enunciations about the genus or species to which that individual belongs. Thus, we are capable of true *scientia* even though we gather our data from individuals.

[14] If our intellection began with explicit and formal knowledge of *ens*, we would expect children just learning to speak to turn to their mothers and say "Entity!" instead of "Mommy!" Since that is obviously not the case, we know that our first knowledge of *ens* is implicit.

[15] We say that knowledge gained formally and explicitly is obtained *actu signato*, through an act (of apprehension or judgment) that is explicitly designated for this purpose.

presenting itself painfully to me is clearly something that *is*, and it cannot possibly *not* be. This judgment that we make, also *actu exercito*, whenever we are presented with an entity is called the *principle of non-contradiction*, the very first principle of reason.

As it turns out, the very exercise of our intellect reveals a whole store of first principles: the principle of causality (namely, that every entity presenting itself to us is on some level an effect that must have a cause), the principle that the whole is greater than the part, the principle of identity (that an entity is itself, and not some other entity), and so on. Aquinas argues that this set of principles that our reason, even before our formal study of logic, infallibly sets forth constitutes an *habitus* that he calls the *habitus primorum principiorum.*[16] Since this *habitus* derives its efficacy from the very exercise of our reason, it is present in all human beings, infallibly, and without exception.

It is these principles that make logic possible. In fact, if you recall, I said that it was defensible to call reason itself a kind of "natural logic." This is true because of the first principles of reason.

3.5 Putting All the Pieces Together

Therefore, to summarize what we have learned so far:

- Logic is an instrumentally speculative and modally practical science. We saw in the first chapter how logic fulfills the characteristics of both a *science* and *art*, and, indeed can be characterized as both a speculative and a practical science. It can be all three at once because, unique among all intellectual *habitus*, its very object of study, the works of reason, is also what it produces, and it is also the means by which it reaches the truth. Because logic is the means par excellence by which all other sciences attain true knowledge of the world outside the soul, but does not seek that kind of knowledge in itself we say that it is *instrumentally speculative*. Because logic helps other sciences obtain that knowledge by the correct construction of relations between the works of reason, we say that logic is *modally practical*.

- The material object of logic is the works of reason. Logic concerns itself as such with concepts, enunciations, and arguments: the fruit of the operations of our intellect. It bears reiterating: *logic studies the* works* of reason, not the operations of reason*.

[16] Rather confusingly, Aquinas also gives this *habitus* the name of *intellectus*. Thus, it whenever the term *intellectus* appears in his works, it is very important to understand which meaning he is giving the term at a given moment.

- The formal object *quod* of logic—that specific aspect of the material object that logic is concerned with—is the relations of reason that can be constructed between second-order intentions. Logic looks reflexively on the works of reason, and tells us how they are to be ordered if we are to acquire true and certain knowledge. It is all the other sciences—metaphysics, mathematics, physics, chemistry, biology, economics, astronomy, and so on—that deal with *first* intentions.

- Finally, what permits us to use our reason in a logical fashion (what we called natural logic), and then to study the works of reason (what we called scientific logic), is the first principles of natural reason. Thus, we say that the formal object *quo* of logic is the light of the first principles.

In short,

> Logic is the instrumentally speculative and modally practical science whose material object is the works of reason, whose formal object *quod* is the second-intention relations of reason constructed between those works, and whose formal object *quo* is the light of the first principles of reason.

3.6 The Divisions of Logic

Now that we have established what logic actually is, I can say something about how the course is organized. The works of reason—concepts, enunciations, and arguments—constitute the material object of logic, as we saw, and for this reason we divide classical logic into three main chapters: the logic of the concept, the logic of the enunciation, and the logic of the argument or syllogism.

Getting to the truth is always the ultimate goal of logic. However, as we will see in the course, a large part of our work in logic involves what we will call understanding the *formal conditions* for obtaining the truth: that is, if we presuppose that we are already starting with sound principles and true knowledge, we can understand how to proceed to *new* knowledge starting from the old. We call this aspect of logic *formal logic* or *minor logic*: the part of logic that studies the *correctness* or *validity* of the relations of reason we construct.

However, studying that aspect of logic alone is incomplete. To illustrate the problem, consider the following syllogism:

> All animals have wings.
> All Frogs are animals.
> Therefore, all frogs have wings.

The syllogism is *formally* perfectly correct. As you will learn in the logic of the argument, the middle term, *animal*, is universal at least once, which is sufficient to join the major and minor premise. In short, it is formally a perfect syllogism in BARBARA. However, the conclusion is evidently false. The reason, of course, is that the major premise ("All animals have wings") is false. (In fact, only *some* animals have wings, but not all.) Thus, as you can see, besides formal correctness, there is an additional condition to reaching a true conclusion: both premises must themselves be true. The conditions under which we can guarantee that *formally* correct reasoning is not merely valid but leads to *true conclusions* are called the *material conditions* for truth, and their study is called *material logic* or *major logic*.

As my example makes clear, it is easy to apply the division between material and formal logic to the logic of the argument. Does it apply equally well to the other parts of logic? For the logic of the concept, it does in a way. concepts are called *universals*,[17] because through one concept, we are able to know many concrete individuals. We can, therefore two aspects of this universality: there is the *manner* in which the concept is attributed universally to its subject, which we call *predicability*[18], and there are the *kinds of objects* (or *contents*) that the concept can refer to.[19] In a sense, the latter aspect constitutes the "material" conditions for the truth of the concept, and thus by analogy, the former constitutes the "formal" conditions, because it leaves aside the question of *what* the concept is referring to. Thus, with the logic of the concept, the division between formal and material logic is somewhat justifiable—even though the criterion for dividing is rather different.

It is much more problematic to apply this division to the logic of the enunciation. As we have already discussed, when we attribute a predicate to a subject (or separate it from that subject), we can do so either mediately—that is, with the help of other enunciations that form our premises—or immediately—that is, by understanding it directly either from the first principles of reason or from experience. Of course, we obtain mediated knowledge through argument, and thus, and we know that the logic of the argument is rightly divided between formal and material logic. What is left is *immediate* knowledge of a real object: however, by its very nature, immediate knowledge is not subject to questions about conditions for truth: it is already *known* to be true.

[17]Traditionally, it is said that etymologically the word "universal" comes from the Latin phrase *unum versus alia*, "one toward (many) others."

[18]You will learn in the course that there are five such ways, or predicables: the genus, the species, the difference, the property, and the (predicable) accident.

[19]You will learn in the course that these types of objects include the transcendentals (the properties of all entities), the categories, and the post-predicaments.

Conclusion

As good logicians, we wished to ask ourselves, with precision and rigor, "What precisely is logic?" We are now in a much better position to answer that question, having explored the place of logic among the other arts and sciences, and even explored exactly how it perfects the intellect as an intellectual *habitus*. You should now be in a position to explain the definition of logic, and especially what its object of study is: the works of reason as regards the relations that we construct between second-order intentions, in the light the first principles of reason. You should now be better prepared to explore that object, in the three great divisions of this science: the logic of the concept, the logic of the enunciation, and the logic of the argument.